

ZS-F33

The Conditional UV Reduction of the Z-Spin Odd Three-Form: Wilson Phase–Flux Transgression, Complete Cellular-Seam Classification, and the Charge-Unit Obstruction to the Absolute Vacuum Scale

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Verification: 31 registered results, with three load-bearing statuses lowered to match what is actually proven. The verification suite is reclassified into 19 ASSERT-COMPUTED checks (which can fail — incidence, Betti, the full Δ_1 spectrum, the complete 48-automorphism / 20-involution cellular-seam classification with its finite torsion image, the Wilson polar-line phase), 4 IDENTITY-REPORTED checks (definitional identities such as $q = \lambda^*$), 1 IMPORTED-PROVEN theorem (Cauchy uniqueness), and 3 STRUCTURAL-ASSUMPTION declarations (kept out of the pass count). New results: the even-dimensional seam-parity counterexample ($\ln T = \ln 4$), the Wilson polar-line derivation of $e^{i\omega}$, the flux-gluing exponential uniqueness $\sigma = e^x$, and the Charge-Unit Obstruction | Zero new fitted parameters | $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**

§0. Abstract

This is the revised UV-reduction sequel to ZS-F32, consolidating the cosmological hand-off scoped as ZS-A31. The title is deliberately weakened from “UV completion”: we do not specify the odd gauge group, representation, charge lattice, or full BV master action, so what is delivered is a *conditional reduction* — a localization of the odd vacuum carrier and a decomposition of its obstructions — not a completion. F32 reduced the absolute scale to one susceptibility through $\rho_{\Lambda,Z} = \frac{1}{2} \chi \cdot \omega^2$, with $\omega = \arg \lambda^* = 2.2592495540$ and $\omega^2/2 = 2.5521042734$, leaving $\theta_Z = \omega$, $\beta_{\pm} = 0$, and χ open. We do not close B3.

Five results, each split into its honest components. (i) **Equivariant-torsion localization** (F33.1): the standard closed even-dimensional Ray–Singer carrier is excluded (IMPORTED-PROVEN), its application to the 2D Z-sector is DERIVED-CONDITIONAL, and *odd-dimensional uniqueness is now CLOSED-NEGATIVE* — the 2D seam-parity-restricted determinant torsion $\ln T = \ln 4 \neq 0$ is an explicit counterexample, so a nonzero odd functional does not force an odd-dimensional carrier. (ii) **Wilson phase–flux transgression** (F33.2): the holonomy $\theta_Z = \omega$ is DERIVED-under-WPL, repairing a v1.6 error. A flat connection has $c_1 = 0$ and cannot be the degree-1 theta line, so v1.7 *separates* a flat Wilson polar-line L_W ($c_1 = 0$, holonomy $e^{i\omega} = \lambda^*/|\lambda^*|$ read from the polar unitary part of the linearized mediation operator) from the theta line Θ_Z ($c_1 = 1$, fibre flux). The Koenigs quotient $\mathbb{C}^*/\langle \lambda^* \rangle$ IS an elliptic curve (PROVEN) and a natural fibre candidate, but the physical fibre need only carry a normalized degree-1 class, so “ $Z = E_{\lambda^*}$ ” is DERIVED-CONDITIONAL and $q = \lambda^*$ is a definitional identity. (iii) An **eighth-power scaling classification** (F33.3): the dimensional law is DERIVED; $E^* = v$ is a rejected benchmark; the blind benchmark $\rho_{\Lambda}^{1/4} = 2.48 \times 10^{-5}$ eV remains 90× short. (iv) A **BRST-Only Isolation No-Go** (F33.4A) — gauge invariance and seam parity alone *cannot* forbid the

mixed operators $F^{-2}R$, $F^{-2}F^{\pm 2}$. The v1.6 claim that $\int F_4$ is a nontrivial local $H^{04}(s|d)$ class is *retracted* (it is d-exact); v1.7 instead writes a two-four-form sequestering action with the corpus $F^{\pm}$, whose abelian-ghost-tower BV master equation reduces to $d^2 = 0$ (DERIVED-CONDITIONAL), and pins the gluing functions by the exponential-uniqueness theorem $\sigma = e^{\chi}$. (v) The spectral no-go is upgraded to a **complete cellular-seam classification** (F33.5): 48 cellular automorphisms, 20 involutions, seam-odd torsion image $\{-3, 0, 1, 3/2, 2, 5/2\} \cdot \ln 2$, disjoint from the back-solved target — exact for all cellular involutions on this complex. The frozen top-form gives $w = -1$ (IMPORTED-PROVEN); the dynamical-w branch is CLOSED-NEGATIVE under the minimal action. Finally, the absolute value: the three registered routes are exhausted (CLOSED-NEGATIVE-under-R1–R3), and the **Charge-Unit Obstruction** explains why — flux integrality fixes the flux number k but not the dimensionful unit e^{-2}/Z , so (A, Q) and topology alone cannot determine χ . The absolute scale in a broader UV theory is OPEN. This is the honest content of conditional reduction: the structure is derived, the absolute scale is provably beyond (A, Q) plus topology.

The computational center of the paper is F33.5, *now a complete classification*. We construct the full truncated-octahedron BCC T^3 quotient explicitly — incidence matrices ∂_1, ∂_2 reproducing $\Delta_1 = \{0^3, 4^3, 6^2, 8^3, 12^1\}$ and Betti $(1, 3, 3, 1)$ — and enumerate its entire signed cellular automorphism group: 48 automorphisms ($= |O_h|$), 20 order-2 seam involutions. For every involution we compute the seam-odd determinant torsion; the image is the finite signed half-lattice $\{-3, 0, 1, 3/4, 2, 5/2\} \cdot \ln 2$, and the back-solved target lies in none of these sets (nor in the non-negative lattice the ordinary subdeterminants occupy, whose nearest point is 8.148). This corrects the v1.6 over-scope: the equivariant torsion is a *signed* half-lattice (the canonical value is $-3 \ln 2 < 0$), not the non-negative lattice claimed there. The spectral / cellular-torsion route to the vacuum scale is **CLOSED-NEGATIVE** — a complete no-go for all cellular seam involutions on this fixed complex (it does not claim to exclude refined torsions or other complexes). $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

Epistemic Status Legend

STATUS	DEFINITION
PROVEN / IMPORTED-PROVEN	Complete proof from standard mathematics, in this paper or in the cited peer-reviewed literature.
PROVEN-under-product	Proved given an explicitly stated product/fibration ansatz; the ansatz is conditional.
PROVEN-under-canonical-action	Proved given the canonical (minimal Abelian compact) action; nonminimal extensions are separate.
DERIVED	Follows from Z-Spin structure plus standard mathematics; no new fitted parameter.
DERIVED-CONDITIONAL	Derived conditional on a stated hypothesis not closed here.
DERIVED-under-WPL	Derived conditional on the Wilson Polar-Line identification (a stated upstream-operator hypothesis).
TARGET-DERIVED	Derived from a stated gluing/uniqueness axiom that is itself the deliberate target of the theorem.
STRUCTURAL-CORRESPONDENCE	A structural match between a corpus object and an external construction, short of an identity.
IMPORTED-MOTIVATED / IMPORTED-BENCHMARK	An external mechanism invoked as motivation, or an external value used as a benchmark, not a Z-Spin prediction.
HYPOTHESIS-strong	A well-motivated conjecture with a named closure route, not yet derived.
NO-GO	A registered impossibility: the stated structure cannot achieve the stated goal.

STATUS	DEFINITION
CLOSED-NEGATIVE	A registered route or equality resolved in the negative (scope stated).
CONFIRMED-OPEN	A closure attempt was made and explicitly failed; the gap is confirmed undecidable with the present tools.
OPEN / OPEN-executable / NON-CLAIM	A recognized gap; “executable” when an algorithm is supplied; NON-CLAIM when the paper deliberately asserts nothing.
LOCKED	Core constant fixed upstream; immutable downstream.

Claim Ledger (31 results)

v1.7 lowers three load-bearing statuses to match what is actually proven ($\theta_Z = \omega$, the Kaloper–Padilla embedding, and the spectral no-go scope), retracts one incorrect cohomology claim, and registers four new results (a parity counterexample, a Wilson polar-line holonomy, a flux-gluing uniqueness theorem, and the Charge-Unit Obstruction). The “Status” column is the v1.7 assignment.

Tag	Result	Status
F33.1A	Standard closed even-dim Ray–Singer torsion is trivial ($T_{RS} = 1$)	IMPORTED-PROVEN
F33.1B	A standard 2D Z-sector carrier is excluded	DERIVED-CONDITIONAL
F33.1C	Odd-dimensional carrier uniqueness — refuted by counterexample	CLOSED-NEGATIVE
F33.1C*	2D seam-parity-restricted determinant torsion $\ln T_- = \ln 4 \neq 0$	PROVEN
F33.2A	Differential-character transgression $\pi!(\hat{a} \cup \hat{u}Z) = \hat{a}$	PROVEN-under-product
F33.2B	Wilson polar-line holonomy $\text{Hol}(L_W) = e^{i\omega}$ on the seam eigenline	DERIVED-under-WPL
F33.2B-i	Koenigs quotient $\mathbb{C}^*/\langle \lambda^* \rangle$ is an elliptic curve	PROVEN
F33.2B-ii	Wilson–Koenigs orbit space $\mathbb{C}^*/\langle \lambda^* \rangle = E_{\lambda^*}$ is the F0 Wilson orbit space	DERIVED-under-Orbit-Equivalence
F33.2B-iii	Theta line on $\text{Jac}(Z)$ has $c_1 = 1$ (fibre flux normalization, distinct from L_W)	PROVEN
F33.2B-iv	Nome $q = e^{2\pi i \tau} = \lambda^*$ (definitional identity)	PROVEN (identity)
F33.2C	$\theta_Z = \omega$ (phase from L_W , flux from Θ_Z)	DERIVED-under-WPL
F33.3A'-math	Dimension consistency $2(d-1)=d(d-1)/2$ ($d>1$) $\Rightarrow d=4, X=3, Y=6$	PROVEN
F33.3A'-phys	Y-sector carries both $X \otimes \mathbb{R}\mathbb{C}$ and $\Lambda^2(\mathbb{R}^{1,3})$ representations ($Q=11$ cross-check)	DERIVED-CONDITIONAL
F33.3A	Eighth-power dimensional law $\rho \sim E^{*8}/\tilde{M}P^4$ admissible	DERIVED
F33.3B	KV eighth-power scaling within a specific q-theory model	IMPORTED-PROVEN-within-model
F33.3C	Odd-sector scale $E^* = v$	BENCHMARK (rejected as value closure)
F33.4A	BRST + seam parity cannot forbid F^2R, F^2F^2	NO-GO
F33.4B	$\int F_4$ as a nontrivial local $H^{04}(s d)$ class	RETRACTED $\rightarrow$ OPEN-GLOBAL
F33.4B-i	Two-four-form BV master action $(S_{BV}, S_{BV}) = 0$ (abelian ghost tower)	DERIVED-CONDITIONAL
F33.4B-ii	Flux-gluing exponential uniqueness $\sigma(x+y)=\sigma(x)\sigma(y) \Rightarrow \sigma = e^x$	TARGET-DERIVED
F33.4B-iii	Z-Spin $\leftrightarrow$ local-sequestering embedding	DERIVED-CONDITIONAL
F33.5a	Explicit TO/BCC T^3 quotient; $\Delta_1 = \{0^3, 4^3, 6^2, 8^3, 12^1\}$, Betti (1,3,3,1)	PROVEN
F33.5b	Complete cellular-seam classification: 48 automorphisms, 20 involutions	PROVEN

Tag	Result	Status
F33.5c	Seam-odd torsion image = $\{-3,0,1,3/2,2,5/2\} \cdot \ln 2$; target $\notin$ image	PROVEN
F33.6a	Canonical compact 3-form: $E_k(\theta) = \frac{1}{2}\chi_-(\theta+2\pi k)^2$, so $b_2\Box = 0$	PROVEN-under-canonical-action
F33.7A	Frozen top-form cosmology $w = -1$	IMPORTED-PROVEN
F33.7B	Dynamical $w(z) \neq -1$ in the minimal 3-form action	CLOSED-NEGATIVE
F33.8a	B3 value via three registered routes (spectral, $E^*=v$, KP-tautology)	CLOSED-NEGATIVE-under-R1–R3
F33.8b	Charge-Unit Obstruction: topology fixes flux number, not flux unit	PROVEN-under-minimal-EFT
F33.8c	Absolute χ_- in a broader UV theory (closed via F33.8D route)	OPEN
F33.8D	Charge-unit reduction $\chi_- = e^{-2}/(4\pi^2 Z_-)$ under a rank-2 internal-cycle ansatz (ZS-F34)	DERIVED-CONDITIONAL (under uplift ansatz)

Cross-Version Consistency (ZS-F32 / A28 / A29)

v1.0 contained two cross-version conflicts, now resolved.

Upstream result	v1.0 tension	v1.1 resolution
A28 v1.2: chiral-Pfaffian $\frac{1}{2}$ not forced; spectral route retired	v1.0/v1.1 reopened a spectral determinant	F33.5 (v1.2) now COMPUTES the canonical equivariant torsion explicitly and proves 8.190 is not in the achievable lattice $\mathbb{Z} \ln 2 \oplus \mathbb{Z} \ln 3$ — strengthening A28: the spectral route is closed-negative by computation, not by assumption.
A28 (field half): on-shell top-form $\Rightarrow w = -1$ (IMPORTED-PROVEN); A29: i-tetration orbit barely evolves over DESI redshifts	v1.0 asserted $w = -1 + \delta w$ “because the KV remnant is not frozen”	F33.7 keeps $w = -1$ (Branch A) as the on-shell result; a dynamical δw requires a new kinetic $K(q)$ and is deferred to the Outlook (Branch B, OPEN, breaks zero-parameter).

§1. Introduction

1.1 From completion to conditional reduction

ZS-F32 reduced the absolute vacuum scale to a UV odd-sector susceptibility and isolated the residual in three layers: $\theta_Z = \omega$ (DERIVED-CONDITIONAL), $\beta_{+-} = 0$ (OPEN), and $\chi_-(\mathbf{A}, \mathbf{Q}, \mathbf{M}_p)$ (OPEN). A genuine UV completion would require specifying the tuple $(G_-, R_-, \Gamma_{\text{charge}}, S_{BV}, H^{04}(s|d), \chi_-, C_{\text{odd}^{\wedge}sp})$: odd gauge group, matter representation, charge lattice, BV master action, counterterm cohomology, susceptibility and determinant. We specify none of the first four. v1.1 therefore claims only a conditional reduction, and reclassifies every result accordingly (Claim Ledger).

The cosmological consolidation of A31 carries a hard constraint we obey: a unified paper that also compares to data must not use the observed ρ_Λ as an input. The cosmological section is therefore blind, and the epistemic ceiling of the paper is set by its weakest load-bearing node, $C_{\text{odd}^{\wedge}sp}$.

1.2 Locked inputs

$$\mathbf{A} = \delta_x \cdot \delta_y = (5/19)(7/23) = 35/437, \quad \mathbf{Q} = 11, \quad (\dim Z, \dim X, \dim Y) = (2, 3, 6).$$

$$z^* = i^{(z^*)} \text{ (the i-tetration fixed point), } \lambda^* = (i\pi/2) z^*, \quad \kappa = -\ln|\lambda^*| = 0.1148346, \quad \omega = \arg \lambda^* = 2.2592495540.$$

§2. Equivariant-Torsion Localization of the Odd Carrier (F33.1)

v1.0 stated a single DERIVED localization theorem. Three gaps force a split into an imported theorem, a conditional application, and an open uniqueness gate.

2.1 F33.1A — Standard even-dimensional torsion no-go [IMPORTED-PROVEN]

Theorem F33.1A. If (Z, g, E, ∇_E) is compact, closed, oriented and *even-dimensional*, with E a flat unitary bundle, then the Ray–Singer analytic torsion is trivial: $T_{RS}(Z, E) = 1$, i.e. $\ln T_{RS} = 0$ [1–3]. (Note: the correct statement is triviality, $T_{RS} = 1$; “the torsion vanishes” is loose.) Since $\dim Z = 2$ (even, ZS-F5, LOCKED), Z is in the scope of this theorem.

2.2 F33.1B — Application to the Z carrier [DERIVED-CONDITIONAL]

Proposition F33.1B. Conditional on the Z -sector being realized as a closed oriented 2-manifold carrying a flat unitary local system in the scope of F33.1A, a *standard* Ray–Singer carrier for the odd vacuum functional on Z is excluded. The gauge-mode channel is also excluded: the J_Z -odd subspace of the register is one-dimensional (the slot-1 gauge direction; ZS-F2 §11.8), and this gauge mode is projected out of the physical Hilbert space, so with $\rho_Z = 0$ PROVEN (ZS-F9, the Z -Spin form of “equilibrium vacuum does not gravitate,” $\rho_V(q_0) = 0$ [14]) it sources no curvature. (**Correction to v1.0:** we drop the identification $\text{ind}(D_Z) = \beta_0(Z) = 1$. The constant harmonic mode counted by β_0 is J_Z -even, so $H^0_-(Z) = 0$; the odd gauge channel is the slot-1 direction, not a β_0 mode.)

2.3 F33.1C — Odd-dimensional uniqueness: refuted by counterexample [CLOSED-NEGATIVE]

The historical gate F33.1C. Triviality of the *standard* closed even-dimensional torsion (F33.1A) does not by itself exclude (a) torsion of a complex with boundary, (b) equivariant torsion, (c) refined / Cappell–Miller torsion, (d) a non-unitary local system, (e) mapping-cone torsion, or (f) orbifold / singular torsion. v1.0–v1.5 therefore left open whether an odd-dimensional carrier is *forced*. v1.7 settles this in the negative: the seam-parity-restricted determinant torsion is computed on an explicit even-dimensional carrier and is nonzero. The relevant object is

$$\ln T_- = \frac{1}{2} \sum_p (-1)^p p \ln \det'(\Delta_p|_{\text{im } P_-}), \quad P_{p,-} = (1 - J_p)/2,$$

which is well-defined after the chain-involution conditions $J_p^2 = I$ and $J_{p+1} d_p = d_p J_p$ (verified in §6 on the full complex). On the 2D even carrier this evaluates to $\ln 4 \neq 0$ (§2.3.1), so the uniqueness claim is *false*, not open: an odd-dimensional carrier is *not* forced. **Note (terminology):** $C_M^{\wedge \text{sp}} = \ln \det \Delta_{1, \text{coexact}}$ is a single Hodge-block log-determinant; calling it “the Ray–Singer torsion side” of a Cheeger–Müller pair requires the remaining p -form determinants to combine into the alternating product, which §6 treats as an executable computation rather than asserting.

2.3.1 The v1.6 closure test and its refutation

Now that the Z -sector is fixed as the 2D Koenigs torus (§3.2.2), a specific closure suggests itself: the seam involution is the central inversion $-I$, which is orientation-*preserving* in even dimensions ($(-1)^2 = +1$) and orientation-*reversing* in odd ($(-1)^3 = -1$). One might hope that Hodge duality then forces the seam-odd equivariant torsion to *vanish* on the 2D even carrier — excluding it — while permitting it on the 3D odd carrier, thereby forcing odd-dimensionality. v1.6 tests this explicitly, and the computation *refutes* it.

On a 2D T^2 (4×4 cubical complex) the *full* complex does satisfy Poincaré duality, $\log \det' \Delta_0 = \log \det' \Delta_2 = 20.337$; but *within the τ -odd subcomplex the duality fails*, $\log \det' \Delta_0|_- = 7.742 \neq \log \det' \Delta_2|_- = 10.515$, under every chain-map sign convention tested (Appendix A, computation ET). The seam-odd torsion is therefore

$$\ln T\text{-(2D)} = \frac{1}{2} \sum_p (-1)^p p \cdot \ln \det'(\Delta_p|_-) = \ln 4 \neq 0, \quad \text{while} \quad \ln T\text{-(3D)} = 6 \ln 2.$$

So the even-dimensional seam-odd equivariant torsion is *genuinely nonzero*: the 2D even carrier is *not* excluded, and the orientation-factor argument is insufficient because the τ -odd subcomplex does not inherit Poincaré self-duality (the cellular involution and its Hodge-dual carry an unavoidable relative sign within the odd projection). This confirms the v1.1 “Problem B” observation directly. **Verdict:** the original claim — that a nonzero odd vacuum functional *requires* an odd-dimensional carrier — is *false*: the 2D even torus is an explicit counterexample. Odd-dimensional uniqueness is therefore CLOSED-NEGATIVE, not merely open. **Terminology:** the quantity computed here is the *seam-parity-restricted analytic determinant torsion* (the determinant of the τ -odd Hodge blocks), which is *not* the trace-weighted equivariant Ray–Singer torsion of the external literature — some definitions of the latter do vanish in even dimensions, so we keep the names distinct. The constructive content is recorded next.

2.3.2 What is established: a finite spectral theorem, not a uniqueness result

Although uniqueness is open, the corpus *assignment* of the two roles is consistent and structurally determined. The register splits as $Q = 11 = 3 (X) + 6 (Y) + 2 (Z)$ (ZS-F18 / F33.3A). The *odd vacuum determinant* lives on the 3D X-ambient — the truncated-octahedron BCC T^3 whose edge Laplacian reproduces $ZS\text{-}Q3 = \{0^3, 4^3, 6^2, 8^3, 12^1\}$ (ZS-M6, §6) — while $\theta Z = \omega$ lives on the 2D Z-sector Koenigs torus through the theta line (§3.2). These are different invariants (Ray–Singer torsion vs determinant-line holonomy) on different sectors (3D vs 2D); the integrity audit (Appendix F) confirms no computation conflates them. The assignment is thus DERIVED-from-structure — the corpus geometry places ZS-Q3 on the X-ambient — even though the abstract claim that the carrier *must* be odd-dimensional remains OPEN. **F33-G1C (revised):** a proof that the seam-odd equivariant torsion vanishes on every even-dimensional carrier would reopen the uniqueness route; a demonstration that ZS-Q3 does not live on a 3D complex would break the assignment.

§3. Transgression and the Dynamical Holonomy Identification (F33.2)

3.1 F33.2A — Differential-character transgression [PROVEN-under-product]

Theorem F33.2A. Let $\hat{a} \in \hat{H}^2(X)$ be a degree-2 differential character (a $U(1)$ connection) and $\hat{u}_Z \in \hat{H}^2(Z)$ a fibre character with $\int_Z \text{curv}(\hat{u}_Z) = 1$, on a fibration $Z \hookrightarrow M \xrightarrow{\pi} X$. Then fibre integration of the differential cup product transgresses to the base:

$$\pi_!(\hat{a} \cup \hat{u}_Z) = \hat{a}, \quad \text{equivalently} \quad \text{Hol}_{Y \times Z}(C_3^-) = \exp(i \oint_{Y \times Z} a_Z), \quad C_3^- = (1/2\pi) a_Z \wedge \Omega_Z.$$

This is the Cheeger–Simons / Bär–Becker fibre-integration theorem [10] refined by the differential cup square $BU(1)_{\text{conn}} \rightarrow B^3U(1)_{\text{conn}}$ [11]. (Normalization corrected from v1.0: the fibre factor is $(1/2\pi) \int_Z \Omega_Z = 1$, not $1/2\pi$.) The non-degeneracy condition (i) of v1.0 stands: a_Z must be pulled back from X , or $a_Z \wedge \Omega_Z$

vanishes on the 2D fibre; ZS-F18 Bridge Four supplies the generation $X = [J_{R1}, J_{R2}]$ but its promotion to a literal bundle is part of the conditional content.

3.2 F33.2B — Wilson polar-line holonomy and the phase–flux separation [DERIVED-under-WPL]

Theorem F33.2A proves the transgression *given* $\oint_{\gamma_Z} a_Z = \omega$. v1.3–v1.6 tried to supply that holonomy by identifying the physical connection a_Z with the Quillen theta-line connection on $\text{Jac}(Z)$. That identification was flawed: a flat $U(1)$ connection on T^2 has zero curvature and hence $c_1 = 0$, whereas the degree-1 theta line has $c_1 = 1$, so a single bundle cannot be both. v1.7 corrects this by *separating* the object that carries the phase from the object that carries the flux normalization.

3.2.1 The phase: a flat Wilson polar-line [DERIVED-under-WPL]

Let W be the linearized mediation (Koopman) operator of the i -tetration at its fixed point, acting on the seam-oriented eigenline L_W by $W\psi_- = \lambda^*\psi_-$ with the multiplier λ^* ($0 < |\lambda^*| < 1$). Its polar decomposition $W = U_W|W|$ has, on that eigenline,

$$U_W \psi_- = (\lambda^*/|\lambda^*|) \psi_- = e^{i\omega} \psi_-, \quad \arg(\lambda^*/|\lambda^*|) = \omega \quad (\text{verified to } 10^{-12}).$$

So L_W carries a *flat* unitary connection $a_W = (\omega/2\pi) d\phi$ with $\oint_{\gamma_W} a_W = \omega$ and $\text{Hol}_{\gamma_W}(L_W) = e^{i\omega}$ — and, being flat, $c_1(L_W) = 0$. This is the **Wilson Polar-Line (WPL)** theorem: the holonomy is the phase of the multiplier, read off from the polar (unitary) part of the upstream operator, with *no* reference to a curvature-carrying bundle. **Status:** DERIVED-under-WPL — conditional only on the identification of W with the corpus's upstream Wilson/Koopman operator and the simplicity of λ^* on the seam eigenline (both checked in Appendix A). The engineered Berry-loop of v1.2 is *demoted*: because the loop angle was set to ω by construction, it demonstrates realizability, not derivation, and is retained only as a realizability check.

3.2.2 The flux: an independent degree-one theta line [PROVEN as mathematics]

Transgression (F33.2A) also needs a normalized integral 2-class on the fibre. This is a *separate* object Θ_Z with

$$c_1(\Theta_Z) = 1, \quad (1/2\pi) \int_Z \Omega_Z = 1,$$

realized by the degree-one theta line on $\text{Jac}(Z)$ (Quillen). The Koenigs quotient $\mathbb{C}^*/\langle \lambda^* \rangle = \mathbb{C}/(2\pi i\mathbb{Z} + (\log \lambda^*)\mathbb{Z})$ is an elliptic curve (PROVEN: a rank-2 lattice quotient of $\mathbb{C}$), with $\tau = (\log \lambda^*)/(2\pi i) = (\omega + i\kappa)/2\pi$, and is the natural fibre candidate; its nome is $q = e^{2\pi i\tau} = \lambda^*$ (a definitional identity, F33.2B-iv). But the transgression does *not* require the physical fibre Z to *be* this torus — only that it carry a normalized degree-one class. Hence “physical $Z = E_{\lambda^*}$ ” is DERIVED-CONDITIONAL, and the $c_1 = 1$ statement is a theorem about Θ_Z alone, kept distinct from the flat L_W .

Total transgression. Cupping the two genuinely distinct objects, the degree-4 differential character $C_3 = a_W \cup u_Z$ on $\gamma_W \times Z$ fibre-integrates to $\pi!(C_3) = a_W$, so

$$\text{Hol}_{\gamma_W \times Z}(C_3) = \text{Hol}_{\gamma_W}(L_W) = e^{i\omega},$$

with L_W (flat, $c_1 = 0$) supplying the dynamical phase and Θ_Z ($c_1 = 1$) supplying the fibre flux normalization. Therefore $\theta_Z = \omega$ is DERIVED-under-WPL — no longer asserted via the false “flat = degree-1” identification. **F33-G2WPL:** if the corpus's upstream Wilson/Koopman operator does not have

λ^* as a simple eigenvalue on the seam eigenline, or its polar phase differs from ω , the holonomy claim is RETRACTED.

Distinctness from the gauge holonomy. This dynamical ω is *not* the gauge holonomy $\oint \omega = A = 0.0801$ fixed by Gauss–Bonnet (ZS-F4): $\omega/A = 28.2$, and the two live on different bundles — the Wilson polar-line of the mediation operator versus the $U(1)$ gauge bundle on the polyhedral defect.

3.2.4 Closing the fibre: the Wilson–Koenigs orbit-space theorem [DERIVED-under-Orbit-Equivalence]

v1.7 left “physical $Z = E_{\lambda^*}$ ” as DERIVED-CONDITIONAL — a natural candidate, not a forced one. v1.8 closes it from the F0 dynamics rather than by fiat. The corpus F0 Wilson operator W (an 11×11 register map) has a 2×2 Z -block that, in the seam basis, is the rotation–scaling

$$W_Z = \begin{bmatrix} \operatorname{Re} \lambda^* & -\operatorname{Im} \lambda^* \\ \operatorname{Im} \lambda^* & \operatorname{Re} \lambda^* \end{bmatrix}, \text{ so in the complex coordinate } w, \quad W_Z : w \mapsto \lambda^* w.$$

We verify $P_Z W P_Z = W_Z$, that its eigenvalues are exactly λ^* , $\bar{\lambda}^*$, and that the seam eigenvector $(1, -i)/\sqrt{2}$ is the λ^* eigenvector — the chirality condition J_Z that selects the holomorphic branch over its conjugate (Appendix A, computation WK). Because $0 < |\lambda^*| < 1$, the iterate group $\mathbb{Z}$ acting by $n : w \mapsto \lambda^{*n} w$ on $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is *free and properly discontinuous*, so the quotient is a smooth complex torus,

$$Z_{\text{orb}} = \mathbb{C}^* / \langle \lambda^* \rangle \cong \mathbb{C} / (2\pi i \mathbb{Z} + (\log \lambda^*) \mathbb{Z}) = E_{\lambda^*}, \quad \tau = (\log \lambda^*) / (2\pi i) = (\omega + i\kappa) / 2\pi, \quad \operatorname{Im} \tau > 0.$$

This much is PROVEN (a rank-2 lattice quotient of $\mathbb{C}^*$): the orbit space of the F0 Wilson dynamics *is* the elliptic curve E_{λ^*} . The remaining bridge — that physical observables are constant on Wilson orbits, $O(w) = O(\lambda^* w)$, so that $Z_{\text{phys}} = Z_{\text{orb}}$ rather than merely maps to it — would follow from a BV–BFV cobordism-gluing factorization $\mathcal{A}_{\text{phys}} = \mathcal{A}(\mathbb{C}^*) / \langle W_Z \rangle$; we state it as an orbit-equivalence postulate. The physical-fibre identification is therefore registered DERIVED-under-Orbit-Equivalence: PROVEN as the orbit space, conditional only on observables being constant on Wilson orbits. **F33-G2orb**: if the F0 Wilson Z -block does not have λ^* as a simple eigenvalue, or the seam chirality does not select the λ^* branch, the orbit-space identification is RETRACTED.

3.2.3 Cross-version consistency [PROVEN identity, not an independent bridge]

The nome identity $q = e^{2\pi i \tau} = e^{\log \lambda^*} = \lambda^*$ ($|q - \lambda^*| < 10^{-30}$) is *definitional*: τ was defined as $(\log \lambda^*) / (2\pi i)$, so $q = \lambda^*$ follows automatically. It is a useful consistency statement — the same λ^* that sets ω and κ is the nome of the candidate fibre torus — but it is *not* independent evidence that the carrier and the i -tetration coincide, and we no longer present it as such. The j -invariant $j(\tau) = 1.06 \times 10^5 + 2.41 \times 10^5 i$ is a determined function of λ^* ; we report it and force no relation to (A, Q) . The 4π /quaternionic closure (ZS-F27) is consistent with the odd-characteristic choice. No cross-version conflict arises.

§4. Eighth-Power Scaling: Classification and Blind Benchmark (F33.3)

Retitled from “Scaling Resolution”: the form is classified and a blind benchmark computed, but the normalization is not resolved.

4.1 F33.3A’ — Bivector–complexification consistency: a second derivation of (3, 6, 2) [math PROVEN]

Before the scaling law, v1.8 records an independent cross-check of the register dimensions that does *not* assume $Q = 11$. Two descriptions of the Y-sector must agree. (a) If Y is the complex/conjugate doubling of the spatial X-sector, then over $\mathbb{R}$ its dimension is $\dim Y = 2 \cdot \dim X = 2(d-1)$ in d spacetime dimensions. (b) If Y carries the Lorentz curvature / electromagnetic *bivector*, then $\dim Y = \dim \Lambda^2(\mathbb{R}^{\wedge\{1,d-1\}}) = C(d, 2) = d(d-1)/2$. Equating the two,

$$2(d-1) = d(d-1)/2 \Rightarrow d = 4 \quad (\text{the unique solution with } d > 1), \quad \text{so } X = 3, Y = 6.$$

Thus $Z = 2$ (ZS-F5) plus the bivector identification *forces* $d = 4$, $X = 3$, $Y = 6$ and recovers $Q = 3 + 6 + 2 = 11$ *without* presupposing it. Four independent routes reach the same 6 (Appendix A, computation Y6): $Y = X \otimes_{\mathbb{R}} \mathbb{C} \Rightarrow 3 \times 2$; $Y = \Lambda^2(\mathbb{R}^{\wedge\{1,3\}}) \Rightarrow C(4,2)$; $Y = \mathfrak{so}(1,3) \Rightarrow 3_J + 3_K$ (rotations + boosts); and $Y = (1,0) \oplus (0,1) \Rightarrow$ the self-dual/anti-self-dual split $E \pm iB$, each $\mathbb{C}^3$. The isomorphism $\Lambda^2(\mathbb{R}^{\wedge\{1,3\}}) \cong \mathfrak{so}(1,3)$ and the complex self-dual decomposition $\Lambda^2_{\mathbb{C}} = \Lambda^2_{+} \oplus \Lambda^2_{-}$ ($\dim_{\mathbb{C}} 3$ each) are standard. **Status (split):** the dimension identity $2(d-1) = d(d-1)/2 \Rightarrow d = 4$ and the representation equivalences are *PROVEN* (F33.3A'-math); identifying the Y-sector *as* carrying both representations is *DERIVED-CONDITIONAL* (F33.3A'-phys; DERIVED if one adopts the corpus ZS-M2/S15 Maxwell–Poynting reading); the agreement with $Q = 11$ is an INDEPENDENT-CROSS-CHECK.

4.2 F33.3A — Dimensional law [DERIVED]

Since $\rho_{\Lambda,Z} = \frac{1}{2}\chi\omega^2$ forces $[\chi] = (\text{energy})^4$, the gravitationally suppressed four-form object

$$\rho \sim E^{*8} / \bar{M}_p^4, \quad [E^{*8}/\bar{M}_p^4] = (\text{energy})^4,$$

is dimensionally admissible and consistent with F32. This is a classification of admissible forms, not a unique law.

4.3 F33.3B — Klinkhamer–Volovik realization, split by what is actually proven

v1.7 marked this IMPORTED-MOTIVATED as a single block. v1.8 splits it, because the external Klinkhamer–Volovik q-theory proves some of it within a model while leaving the rest phenomenological. In specific q-theory models the vacuum energy obeys $\rho_v(t) \propto \kappa M^2(t) H^4(t)$ and, at the electroweak crossover, $\rho_v \sim E_{ew}^8/E_p^4$ — a genuine eighth-power scaling, but with an order-unity coefficient fixed by microphysics and a freeze-out governed by a phenomenological relaxation rate $\dot{\rho}_v = -\Gamma(t)(\rho_v - \rho_{v,0})$. The honest split is:

Claim	Status
Eighth-power scaling within a specific KV q-theory model	IMPORTED-PROVEN-within-model
A law universal to every four-form	NON-CLAIM
Z-Spin F- is the KV vacuum variable q	DERIVED-CONDITIONAL
The exact post-freeze-out coefficient	OPEN-executable

Closing the ZS–q correspondence (raising the third row to DERIVED-CONDITIONAL $\rightarrow$ DERIVED) would require writing the action $S = \int \sqrt{-g} [K(q)R - \varepsilon(q) + L_Y(T, q)]$ with $q = \star F_-$, imposing $\rho_v(q_0) = 0$ and $\chi_{q_0}^{-1} = q_0^2 \varepsilon''(q_0) > 0$, and computing the Y-sector thermal $\kappa_Y(T)$ and the freeze-out kernel $\Gamma_Y(\omega) = (1/2\omega) \text{Im } G^{\wedge}R(\omega)$ from the Y retarded correlator rather than fitting Γ . That program is deferred to the companion ZS-F34; here F33.3B is registered at the split above. **F33-G3B:** any use of the eighth-power law as universal, or of a fitted Γ presented as derived, is void.

4.4 F33.3C — The identification $E^* = v$ [BENCHMARK, rejected as value closure]

The electroweak vev is itself parameter-free (Factorized Determinant Theorem, ZS-S4 §6.12):

$$v = \bar{M}_p \cdot \exp[-(38/9)(11 \ln 2 + \ln 3)] = \bar{M}_p \cdot e^{-36.831} = 245.93 \text{ GeV},$$

but $E^* = v$ is a separate odd-sector identification. Closing it requires an explicit portal

$$S_{\text{portal}} = \int d^4x \sqrt{-g} (\bar{M}_p^2 q_-) O_{EW}^-, \text{ with the coefficient forced to unity,}$$

which is not yet derived. The blind benchmark is

$$\rho_{\Lambda, Z}^{1/4} |_{\text{blind}} = v^2 / \bar{M}_p = 2.48 \times 10^{-5} \text{ eV}, \text{ versus } 2.24 \times 10^{-3} \text{ eV observed } (\times 90 \text{ short; } E^*_{\text{need}} = 2.335 \text{ TeV} \approx 9.5 v).$$

4.5 F33.3D — The $b_0 = 14.25$ route [CLOSED-NEGATIVE]

A confining-odd-sector parametrization $\Lambda_- = \bar{M}_p \exp[-8\pi^2/(b_0 g_-^2)]$ with $g_-^2 = A$ would require $b_0 = 14.25$, but ZS-A26 proves $32\pi^2/(b_Z A) = 2v_{\text{now}}\pi/A = 8 A_{\text{comp}}^{\text{odd}} = 276.6$ is the present epoch; any exponent built to hit it is tautological, and the polyhedral β -rule gives the non-confining Z-sector $b_0 \in \{4/3, 5/3\}$ (A22 barrier B2). The missing piece is

$$\rho_\Lambda = c_{KV} (v^8 / \bar{M}_p^4) H(\omega);$$

without c_{KV} and $H(\omega)$ computed from an action, B3 stays OPEN.

§5. A BRST-Only Isolation No-Go and an IR Stability Bound (F33.4)

This section is the principal revision. v1.0 sought to *prove* perturbative isolation; the correct cohomological conclusion runs the other way.

5.1 F33.4A — BRST-Only Isolation No-Go [NO-GO]

Theorem F33.4A. Three-form gauge invariance and seam parity alone do not forbid the mixed local operators

$$\bar{M}_p^{-2} F^{-2} R \quad (\text{mass dim } 6) \quad \text{and} \quad \bar{M}_p^{-4} F^{-2} F_+^2 \quad (\text{mass dim } 8).$$

Proof sketch. Both $F^{-2}R$ and $F^{-2}F_+^2$ are gauge-invariant local scalars. In the Batalin–Vilkovisky antifield formalism, a gauge-invariant local functional is annihilated by the BRST differential s and is generically not s -exact, so it is a non-trivial element of the ghost-number-zero local cohomology $H^0(s|d)$ that classifies admissible counterterms and consistent deformations [8,9]. BRST removes gauge-noninvariant terms; it does not remove gauge-invariant higher-dimension operators, which are exactly the content of an effective field theory. Seam parity does not help: since F_- is seam-odd, F^{-2} is seam-even, so the mixed operators are parity-allowed (F32.22). Hence the v1.0 target $[F^{-2}R], [F^{-2}F_+^2] \notin H^0(s|d)$ fails without additional structure: the mixing coefficient β_{+-} is not forced to zero by BRST and parity.

This negative result is more valuable than the asserted positive one: it tells any UV completion that an exact isolation symmetry must be supplied by hand (an extra global symmetry, a Ward identity, or a sequestering mechanism), and it cannot be read off from gauge invariance.

5.2 F33.4B — A two-four-form sequestering action and its BV master action [corrected; BV master DERIVED-CONDITIONAL]

Exact zero is the wrong target; phenomenological stability is the right one. With the mixed operators present, the radiative shift of the susceptibility is

$$|\delta\chi_- / \chi_-| \lesssim |c_R| \langle R \rangle / \bar{M}_P^2 + |c_8| \langle F_+^2 \rangle / \bar{M}_P^4 + \dots$$

In the present universe $\langle R \rangle / \bar{M}_P^2 \sim H_0^2 / \bar{M}_P^2 \sim (10^{-61})^2 \sim 10^{-122}$, and the SM $\langle F_+^2 \rangle / \bar{M}_P^4$ is likewise negligible, so even without exact cancellation the fractional shift is far below any observable threshold. The IR isolation needed for cosmology is therefore *radiative, not exact*.

Correspondence, not identity (v1.7). v1.6 claimed the corpus four-form is “precisely” the Kaloper–Padilla sequestering system. That overstates the match: KP local sequestering requires *two* four-forms and two rigid scalars $\Lambda(x)$, $\kappa^2(x)$ with nonlinear gluing functions, whereas $F_- = dA_3^-$ with $\rho_Z = 0$ supplies only one. The honest relation is a STRUCTURAL-CORRESPONDENCE. To make it precise without new species, v1.7 writes the embedding action using the corpus's own $F_\pm$,

$$S_{ZS-seq} = \int \sqrt{-g} \left[\frac{1}{2} \kappa^2(x) R - \Lambda(x) - L_{ZS} \right] + \int \sigma_-(\Lambda / \bar{M}_P^4) F_- + \int \sigma_+(\kappa^2 / \bar{M}_P^2) F_+,$$

with F_- enforcing the vacuum-shift constraint and F_+ the Planck normalization. The two 4-form gauge symmetries make Λ and κ^2 on-shell constants ($\partial\Lambda = \partial\kappa^2 = 0$), and the residual cosmological constant is the ratio of the two fluxes — which, as the external theorem states, is an arbitrary finite integration constant fixed by measurement, not by (A, Q) . The dim-6 and dim-8 mixed cocycles (admissible by F33.4A) contribute, off this background,

$$|\delta\chi_- / \chi_-| \lesssim \langle R \rangle / \bar{M}_P^2 \sim 10^{-121} \quad (\text{dim } 6), \quad \sim 10^{-241} \quad (\text{dim } 8), \quad \text{both radiatively protected.}$$

Retraction of the local-cohomology claim (v1.7). v1.6 asserted that $\int F_4$ is a nontrivial element of the ghost-number-zero local cohomology $H^{0|4}(s|d)$. This is *withdrawn*: locally $F_4 = dA_3$, so $\int F_4$ is d-exact, and local BRST cohomology quotients by $a \sim a + sb + dc$ — a d-exact term is trivial there. A nontrivial flux class lives in global differential cohomology, a different object that local $H(s|d)$ does not certify; the v1.6 verification merely hard-coded the booleans rather than computing them, so the claim is RETRACTED $\rightarrow$ OPEN-GLOBAL. **What is provable.** The correct statement is at the level of the BV master action. Adjoining antifields and the reducible abelian ghost tower $c_{2,\pm}, c_{1,\pm}, c_{0,\pm}$ to S_{ZS-seq} gives the explicit action $S_{BV} = S_{cl} + \sum_\pm \int [A^*_{3,\pm} \wedge dc_{2,\pm} + c^*_{2,\pm} \wedge dc_{1,\pm} + c^*_{1,\pm} \wedge dc_{0,\pm}]$. For this abelian tower the master equation $(S_{BV}, S_{BV}) = 0$ reduces term-by-term to $d^2 = 0$ plus ghost nilpotency, and the topological $\sigma(\Lambda)F_4$ and $F_\pm$ couplings are s-closed, so they do not deform the gauge algebra. The mechanism is that $\sigma(\Lambda)F_4 = d(\sigma A_3) - \sigma'(\Lambda)d\Lambda \wedge A_3$ is *not* a pure boundary term off-shell, and the A_3 equation of motion forces $d\Lambda = 0$. The reduction to $d^2 = 0$ is exhibited, but the full antibracket with the $F_\pm$ couplings is not computed in the verifier, so this is registered DERIVED-CONDITIONAL, not PROVEN (F33.4B-i). **Closing the free function.** The gluing functions $\sigma_\pm$ are a priori arbitrary, which would break zero-parameter status. They are pinned by a uniqueness theorem (F33.4B-ii): if cobordism gluing adds flux labels while multiplying path-integral weights, then $\sigma(x+y) = \sigma(x)\sigma(y)$ with $\sigma(0) = 1$, $\sigma'(0) = 1$, whose unique continuous solution is $\sigma(x) = e^x$ — the same additive-to-multiplicative homomorphism uniqueness as ZS-M1. **Status:** BRST-only isolation NO-GO (F33.4A) retained; $\int F_4$ local class RETRACTED; two-four-form BV master action DERIVED-CONDITIONAL (abelian ghost tower, master equation reduced to $d^2 = 0$; full antibracket deferred) (F33.4B-i); $\sigma = e^x$ TARGET-DERIVED (F33.4B-ii); the Z-Spin $\leftrightarrow$ local-sequestering embedding DERIVED-CONDITIONAL (F33.4B-iii); matter-loop radiative stability IMPORTED-PROVEN + DERIVED-CONDITIONAL on that embedding. **F33-G4 (revised):** a

demonstration that the BV master equation fails, or that gluing does not multiply weights, falsifies the construction.

§6. Complete Cellular-Seam Classification of the Odd Spectral Determinant (F33.5)

Promoted to the center of the paper and, in v1.2, *resolved*. ZS-A28 retired the naive spectral route (the chiral-Pfaffian $\frac{1}{2}$ is not forced); we now construct the full chain-level object explicitly with a canonical cellular seam involution — no half-weight imposed — and compute its equivariant torsion. The result closes the spectral route to the vacuum scale in the negative.

6.1 The target, and why 8.190 is back-solved

Closure of B3 via §4 would require $C_{\text{odd}}^{\text{sp}} = 34.58/(38/9) = 8.190$, equivalently $E^* = \sqrt{(\rho_{\text{obs}} \bar{M}_p)} = 2.335$ TeV. This value is back-solved from the observation and is a falsification target. The corpus-native log-determinants bracket it but (corrected from v1.0) no simple combination hits it: the nearest half-weight $11 \ln 2 + \frac{1}{2} \ln 3 = 8.174$, the arithmetic mean 8.233, the geometric mean 8.218 (v1.0 wrongly called the last two both 8.233), and the coexact spectrum is $\{8^3, 12^1\}$.

Table 1. Corpus-native log-determinants vs. the back-solved target.

Object	eigenvalues	$\ln \det$ (weight)	value
exact block (dim 5)	$4^3, 6^2$	$8 \ln 2 + 2 \ln 3$	7.742
coexact = C_M^{sp} (dim 4)	$8^3, 12^1$	$11 \ln 2 + \ln 3$	8.723
target (back-solved)	—	not an integer $\ln$ -lattice point	8.190

6.2 The explicit truncated-octahedron BCC T^3 quotient

We build the quotient complex literally. The truncated octahedron (the BCC Voronoi cell) has 24 vertices at the permutations of $(0, \pm 1, \pm 2)$, 36 edges, and 14 faces (6 squares + 8 hexagons). Quotienting by the BCC translation lattice $\Lambda = \langle (2,2,2), (4,0,0), (0,4,0) \rangle$ (covolume 32 = cell volume) identifies vertices 4-to-1, edges 3-to-1, faces 2-to-1, giving a CW complex on T^3 with

$$(V, E, F, C) = (6, 12, 7, 1), \quad \chi = 0, \quad \text{Betti } (b_0, b_1, b_2, b_3) = (1, 3, 3, 1).$$

The oriented incidence matrices ∂_1 (6×12) and ∂_2 (12×7) satisfy $\partial_1 \partial_2 = 0$ (verified), $\text{rank } \partial_1 = 5$, $\text{rank } \partial_2 = 4$, and reproduce the corpus ZS-Q3 edge spectrum exactly:

$$\text{spec } \Delta_1 = \text{spec}(\partial_1^T \partial_1 + \partial_2 \partial_2^T) = \{0^3, 4^3, 6^2, 8^3, 12^1\},$$

with the octahedral vertex Laplacian $\Delta_0 = \partial_1 \partial_1^T$ giving the exact block $\{4^3, 6^2\}$ ($\det 7.742$) and $\Delta_2 = \partial_2^T \partial_2$ giving the coexact block $\{8^3, 12^1\}$ ($\det 8.723 = C_M^{\text{sp}}$). This is the concrete realization the reviewer's Step 1 required, and it closes the v1.1 gap (a): ∂_2 is now in hand.

6.3 The canonical central-inversion seam involution and its torsion

The truncated octahedron is centrally symmetric, so the central inversion $\sigma : v \mapsto -v$ descends to the quotient as a chain involution J_p on every degree. We verify the full package numerically (Appendix A, computation TO):

$$J_0^2 = J_1^2 = J_2^2 = I, \quad J_0 \partial_1 = \partial_1 J_1, \quad J_1 \partial_2 = \partial_2 J_2, \quad [J_p, \Delta_p] = 0 \quad (\text{all } p).$$

This is a genuine cellular seam involution on the *full* complex — not only the 1-skeleton of v1.1 — closing gaps (b)/(c). Its seam-odd (J-odd) blocks are

$$\Delta_0|_ = \{4^3\}, \quad \Delta_1|_ = \{4^3\}, \quad \Delta_2|_ = \emptyset, \quad \Rightarrow \quad \ln T_- = \frac{1}{2} \sum_p (-1)^p p \ln \det'(\Delta_p|_-) = -3 \ln 2 = -2.079,$$

with each nonzero seam-odd determinant equal to $3 \ln 4 = 4.159$. (The seam-odd subcomplex carries the orientation-odd b_3 volume mode, handled by the standard $\det'$ zero-mode subtraction.) The canonical seam-odd torsion $-2.079 = -3 \ln 2$ is one point of the finite image classified in §6.4; none of the constituent determinants (4.159) is 8.190 either.

6.4 The complete cellular-seam classification

v1.6 over-stated the closure as a “non-negative integer lattice” no-go. That was both too weak and self-contradictory: the equivariant torsion $\ln T_- = \frac{1}{2} \sum_p (-1)^p p \ln \det'(\Delta_p|_-)$ carries an alternating sign, a degree weight p , and a factor $\frac{1}{2}$, so its values lie in a *signed half-lattice* — and the canonical value $-3 \ln 2$ is itself negative, hence *not* in $\mathbb{Z}_{\geq 0} \ln 2 \oplus \mathbb{Z}_{\geq 0} \ln 3$. v1.7 replaces the claim with an exact enumeration. We compute the full signed cellular automorphism group of the quotient complex (Appendix A, computation CLS):

$$|\text{Aut}_{\text{cell}}(C.)| = 48 = |O_h|, \quad \#\{J : J^2 = I\} = 20 \quad (19 \text{ nontrivial involutions}),$$

each verified to satisfy $J_p^2 = I$ and the chain-map conditions $J_0 \partial_1 = \partial_1 J_1$, $J_1 \partial_2 = \partial_2 J_2$. For every one of the 20 seam involutions we compute the seam-odd torsion; the image is the finite signed set

$$\mathcal{T}_- = \{-3, 0, 1, 3/2, 2, 5/2\} \cdot \ln 2, \quad \max |\mathcal{T}_-| = 3 \ln 2 = 2.079,$$

and the per-degree determinant images $\ln \det'(\Delta_p|_-)$ are likewise finite sets (Appendix A). The back-solved target 8.190 lies in *none* of these sets — it is more than a factor of three above the largest attainable |torsion|. **Two scopes, kept distinct.** (i) The *ordinary* spectral subdeterminants of ZS-Q3 (the route that would set $C_{\text{odd}}^{\text{sp}}$) do factor over $\{2,3\}$ and lie in the non-negative lattice $\mathbb{Z}_{\geq 0} \ln 2 \oplus \mathbb{Z}_{\geq 0} \ln 3$, whose nearest point to 8.190 is $7 \ln 2 + 3 \ln 3 = 8.148$ — so that route misses. (ii) The *complete seam-odd torsion* over all 20 cellular involutions is the signed half-lattice set $\mathcal{T}_-$ above — which also misses. **Theorem F33.5 (Complete Cellular-Seam Classification) [PROVEN]:** on this fixed BCC T^3 complex, the seam-odd determinant and torsion images are the finite sets above, and the target is disjoint from both. **Scope.** This is a complete no-go for *all cellular seam involutions on this complex*; it does *not* claim to exclude refined / Cappell–Miller torsion or a different complex. **The spectral / cellular-torsion route is CLOSED-NEGATIVE** with the scope now exact.

6.5 What is closed, what remains, and the firewall

Closed: the hypothesis “ p_Λ is a seam-odd spectral/equivariant determinant on the ZS-Q3 complex equal to 8.190” is refuted. **Not closed:** B3 as a whole, since non-spectral mechanisms remain open — the IR radiative-stability / sequestering route (§5.2), a different complex, or genuinely new structure. The register $\leftrightarrow$ CW intertwiner U is now moot for *this* question: whatever U does, the achievable values lie in $\mathcal{L}$ and exclude 8.190. **Firewall.** The verification script `zs_f33_verify_v1_8.py` computes ∂_1 , ∂_2 , J_p , the spectra and the torsion reading *only* incidence data and locked normalizations; the target 8.190 enters solely in a post-computation comparison block. **Status: CLOSED-NEGATIVE (spectral route); B3 OPEN via non-spectral routes.**

§7. Vacuum Functional Form: the Canonical Compact 3-Form Branch (F33.6)

Theorem F33.6 (canonical branch). [PROVEN-under-canonical-action] v1.6 left the higher cumulants $b_2, b_4, \dots$ OPEN and leaned on an SU(3) benchmark. v1.7 closes the leading question by adopting the *canonical* minimal Abelian compact 3-form action,

$$S-[A_3, \theta] = -(1/2\chi_-) \int F_4 \wedge \star F_4 + (\theta/2\pi) \int F_4, \quad (1/2\pi) \int F_4 \in \mathbb{Z}.$$

Integrating out the flux F_4 at fixed integer class k gives an *exactly quadratic* branch functional,

$$E_k(\theta) = \frac{1}{2} \chi_- (\theta + 2\pi k)^2, \quad E(\theta) = \min_{k \in \mathbb{Z}} E_k(\theta),$$

so in the canonical branch the higher cumulants *vanish identically*: $b_2 = b_4 = b_6 = \dots = 0$, and the truncation problem disappears ($\Delta E_-(\theta) = \frac{1}{2} \chi_- \theta^2$ is exact on each branch). **Status:**

PROVEN-under-canonical-action. The corpus large-N / canonical registration of the quadratic law (ZS-F32) is the upstream warrant for this branch. **Non-canonical comparison.** A general nonlinear kinetic function $K(F)$ would reintroduce $b_2, b_4, \dots$ (OPEN, nonminimal extension); the SU(3) benchmark $b_2 = -0.0216$ [5] ($b_2 \omega^2 \approx -11\%$) is one such non-canonical datum, retained only as an IMPORTED-BENCHMARK comparison, not a Z-Spin prediction. **F33-G6:** if the odd sector is shown to require a non-canonical (non-quadratic) kinetic term, the $b_2 \square = 0$ result is confined to the canonical branch and the cosmological use must carry the truncation gate $\varepsilon_N < 1\%$.

§8. Cosmological Hand-off: Frozen Branch and Outlook (F33.7)

8.1 Branch A — Frozen top-form, $w = -1$ [IMPORTED-PROVEN]

On shell a non-propagating four-form in 4D is constant, $T_{\mu\nu}^{(-)} = -\rho_{\Lambda,Z} g_{\mu\nu}$, so $\dot{q} = 0$ and $w = -1$ strictly. ZS-A28 fixes this as the field-theory-half IMPORTED-PROVEN result, and ZS-A29 finds the i-tetration orbit barely evolves across DESI redshifts, freezing $w \approx -1$. The blind Friedmann normalization is

$$3 \bar{M}_p^2 H^2 = \rho_m + \rho_r + \rho_{\Lambda,Z}, \quad H_\Lambda = \sqrt{(\Delta E_-(\omega)/3\bar{M}_p^2)}, \quad \text{with } \Omega_m = 38/121, \Omega_\Lambda = 83/121, \Omega_\Lambda/\Omega_m = 83/38 = 2.184 \approx 2e^\Lambda = 2.167 \text{ (same order, not an identity),}$$

using H_0 only for comparison. On DESI DR2 we state the evidence carefully: flat Λ CDM still describes the data well, while the $w_0 w_a$ (CPL) extension shows a preference whose significance is dataset- and parametrization-dependent — about 3.1σ for DESI+CMB and 2.8σ – 4.2σ once supernovae are added depending on the sample (Pantheon+, Union3, DESY5), with the preferred region $w_0 > -1$, $w_a < 0$ [16]. This is a preference, *not* an exclusion of Λ CDM. Branch A is currently compatible; a persistent evolving-dark-energy signal would create tension. **F33-G8:** a computed ρ_Λ inconsistent with the observed value falsifies the theory (no retuning).

8.2 Branch B — Dynamical $w(z)$ in the minimal action [CLOSED-NEGATIVE]

v1.7 sharpens the Outlook into a no-go. A canonical 4-form has no propagating degree of freedom in four dimensions: its equation of motion is $d\star F_4 = 0$, so $\star F_4 = \text{constant}$, giving $T_{\mu\nu} = -\rho g_{\mu\nu}$ and $w = -1$ identically. Therefore

$w(z) \neq -1$ does not arise from the minimal 3-form action $\Rightarrow$ Branch B is CLOSED-NEGATIVE under that action.

A dynamical branch can be built only by adding a new kinetic term — e.g. $(\partial \star F_4)^2/M^2$ or a scalar kinetic structure $-\frac{1}{2}K(q)(\partial q)^2$ — which introduces a new degree of freedom and a new scale, and (if K is a free function) breaks zero-parameter status. That is a *separate nonminimal extension*, not an open question within F33. **Consequently the cosmology of the present theory is unambiguous:** $w = -1$, fully closed; an evolving-dark-energy signal would be evidence for the nonminimal extension and is a NON-CLAIM of this paper. **Status: CLOSED-NEGATIVE (minimal action); nonminimal dynamical branch is a NON-CLAIM.**

§9. No-Go Consolidation and Blind Prediction (F33.8)

Blind output: $\rho_{\Lambda Z}^{\text{pred}} = \Delta E(\omega) = \Lambda^{-4} H(\omega)$, conditional on the uncomputed $C_{\text{odd}}^{\text{sp}}$. Four no-gos bound why B3 is OPEN, none an impossibility:

(N1) A–Q-only scale no-go (DERIVED, A28): dimensionless $\mathbf{A}, \mathbf{Q}$ cannot fix a scale; the hierarchy selector is $\ln(\rho_{\text{obs}}/\bar{M}_p^4) = -276.6$. (N2) The 276.6 collapse (DERIVED, A26): the candidate exponents are all the present epoch, so $b_Z = 14.25$ is tautological. (N3) Central-shift no-go (PROVEN, F32.15/A26): $i[H+c_0 I, X] = i[H, X]$, so relative/modular tools are blind to the absolute offset; B3 is a missing central normalizer. (N4) BRST-only isolation no-go (this paper, §5): gauge invariance and parity do not forbid the mixed counterterms, so exact $\beta_{+-} = 0$ is not free.

F33-G9: any claim that $C_{\text{odd}}^{\text{sp}}$ or χ_- is parameter-free while using the observed ρ_Λ , a chosen scale, a base-10 coincidence, a fitted half-weight, or a back-solved exponent is void.

9.2 The non-spectral normalizer: a sequestering route [HYPOTHESIS-strong]

v1.2 closed the *spectral* route ($8.190 \notin$ the lattice $\mathcal{L}$). What remains is non-spectral, and the central-shift no-go (N3) names what is needed: because $i[H + c_0 I, X] = i[H, X]$, every relative or modular tool is blind to the absolute offset c_0 , so B3 is a missing *central normalizer* that fixes c_0 by a global, non-spectral condition. The Kaloper–Padilla sequestering constraint is exactly such an object: the rigid Λ carried by the 4-form flux fixes the absolute vacuum energy topologically, outside the spectrum. Identifying it with the corpus four-form (§5.2) gives

$$\rho_\Lambda = (\text{4-form flux constant}) - \langle L_{\text{matter}} \rangle_{\text{cosmic}}, \quad \text{with the KP prediction } \rho_\Lambda \sim \langle \rho_{\text{matter}} \rangle_{\text{today}}.$$

This is structurally consistent with the corpus coincidence $\Omega_\Lambda/\Omega_m = 2e^\Lambda = 2.17 = O(1)$, and it explains both the smallness and the radiative stability of ρ_Λ through one mechanism. **The honest limit:** the 4-form flux is a free boundary condition, so KP alone does not give a parameter-free ρ_Λ ; closing the value requires fixing that flux to the corpus's blind $\rho_\Lambda = \Delta E(\omega)$, which is *not* done here. Hence the route — “B3's central normalizer is the KP sequestering constraint” — is registered HYPOTHESIS-strong (mechanism identified, non-spectral, RG-stable), with the value-closure OPEN. **F33-G10:** any claim that the KP route fixes ρ_Λ parameter-free while leaving the 4-form flux free, or while not matching it to $\Delta E(\omega)$, is void.

9.3 The absolute value: three routes exhausted [CLOSED-NEGATIVE-under-R1–R3]

v1.5 completes the value question. Three independent routes to a parameter-free absolute ρ_Λ have now been examined and all are closed; the table records the terminus.

Route	Attempt	Outcome
Spectral (v1.2)	$C_{\text{odd}}^{\text{sp}}$ from the seam-odd determinant	$8.190 \notin \mathcal{L} \rightarrow \text{CLOSED-NEGATIVE}$
Branch A ($E^*=v$ benchmark)	$\rho_{\Lambda} = \frac{1}{2}\chi\omega^2$ with $E^* = v$ (electroweak benchmark)	$90\times$ short; $\sqrt{90}$ underivable
Branch B: KP residual	$\rho_{\Lambda} = \bar{M}_P^4 e^{(-v_{\text{now}})}$, $v_{\text{now}} = 276.6$ (A26)	right magnitude, tautological

Branch A and the $\sqrt{90}$ enhancement. With the natural corpus scale $E^* = v$ the eighth-power law gives $\rho_{\Lambda}^{1/4} = v^2/\bar{M}_P = 2.48 \times 10^{-5}$ eV, $90\times$ short of 2.24×10^{-3} eV; closing the gap needs $E^*_{\text{need}}/v = \sqrt{90} = 9.49$. We searched for a principled origin and found none. The nearest numerical near-miss is $e^{\omega} = 9.58$ (within $\approx 1\%$), but e^{ω} is the exponential of a *rotation angle* — the natural object attached to ω is $e^{i\omega}$, not e^{ω} — so it carries no mechanism; the other candidates (Q^x , $(1/A)^x$, $|\lambda^*|^x$) all require non-integer fitted exponents. Adopting any of them is exactly the back-solved-exponent numerology F33-G9/G11 forbid, so we reject them. $\sqrt{90}$ is *not* derivable from (A, Q): Branch A is CLOSED-NEGATIVE.

Branch B and the present epoch. The KP cosmic-average residual gives $\rho_{\Lambda} = \bar{M}_P^4 \exp(-v_{\text{now}})$, and with ZS-A26's present epoch $v_{\text{now}} = 276.6$ this reproduces ρ_{obs} to better than a percent — the right magnitude and the $\rho_{\Lambda} \sim \rho_{\text{matter}}$ coincidence, radiatively stable by §5.2. But $v_{\text{now}} = 276.6 = \ln(\bar{M}_P^4/\rho_{\text{obs}})$ is itself *defined* by ρ_{obs} (A26), so Branch B is tautological: it explains why ρ_{Λ} is what it is at the present epoch, not what it must be from (A, Q) alone.

Conclusion (scoped). None of the three *registered routes* R1–R3 produces a parameter-free absolute ρ_{Λ} : the spectral/cellular route is excluded by the complete classification, the $E^*=v$ branch needs an underivable $\sqrt{90}$, and the residual branch is tautological. So the corpus value-programme is CLOSED-NEGATIVE-under-R1–R3 — a confirmed terminus for *those routes*. This is *not* a claim that an absolute ρ_{Λ} is impossible in any UV completion; that broader statement is OPEN (F33.8c). The next subsection shows *why* the routes fail, structurally. **F33-G11 (retained):** any later closure that fits $\sqrt{90}$, 90, or e^{ω} to (A, Q) rather than deriving the enhancement from a stated mechanism is void.

9.4 The Charge-Unit Obstruction [PROVEN-under-minimal-EFT]

The three-route exhaustion is upgraded from an empirical list to a structural theorem. In the canonical compact 3-form EFT (§7), flux integrality fixes the topological class

$$[F_4/2\pi] \in H^4(M; \mathbb{Z}) \Rightarrow k \in \mathbb{Z}, \text{ but } E_k(\theta) = (e^2/2Z_-)(k + \theta/2\pi)^2, \quad \chi_- = e^2/(4\pi^2 Z_-) \propto e^2/Z_-.$$

Topology fixes the flux *number* k , but the energy spacing depends on the membrane charge e - and the kinetic normalization Z_- — the *dimensionful unit* $\chi_- = e^2/(4\pi^2 Z_-)$ itself, which integrality does not constrain. **Theorem F33.8 (Charge-Unit Obstruction) [PROVEN-under-minimal-compact-3-form EFT]:** (A, Q, dim Z) together with flux integrality fix the flux number but not the flux unit, hence cannot determine the absolute susceptibility χ_- . This is the structural reason the routes of §9.3 fail: the spectral route fixes integers (lattice points), the $E^*=v$ route guesses the unit (and misses by $\sqrt{90}$), and the residual route reads the unit off the observation (tautology). It strengthens the A–Q-only no-go (ZS-A28) to the EFT level: topology sets the *number* of flux quanta, never the dimensionful *size* of one. The integer structure of a charge lattice and the physical charge unit are different data (ZS-F32 B3-1). **This is the paper's strongest terminus:** not “we tried three things,” but “topology alone cannot fix the absolute four-form susceptibility.” **F33-G12:** any claim to fix χ_- from (A, Q) and flux integrality alone — without supplying the dimensionful unit e^2/Z_- from additional UV data — is void.

9.5 F33.8D — The internal charge-unit reduction [DERIVED-CONDITIONAL under a rank-2 internal-cycle ansatz; developed in ZS-F34]

The Charge-Unit Obstruction is *not* an impossibility theorem; it names exactly the missing datum — the kinetic normalization Z_- and the minimal membrane charge e_- . v1.8 records the route that supplies them, deferring the full computation to the companion paper. Wrapping the Y-sector as $Y_6 = M_4 \times \Sigma_2$ and expanding the five-form potential $C_5 = A_3^-(x) \wedge \omega_2(y)$ with $\int_{\Sigma_2} \omega_2 = 1$ gives $G_6 = dC_5 = F_4^-(x) \wedge \omega_2(y)$, and the kinetic action reduces to the 4D one with

$$Z_- = (1/g_6^2) \int_{\Sigma_2} \omega_2 \wedge \cdot_2 \omega_2, \quad e_- = N e_6 \int_{\Sigma_2} \omega_2 = N e_6, \quad \chi_- = e_-^2 / (4\pi^2 Z_-) = N^2 e_6^2 g_6^2 / (4\pi^2 V_\Sigma),$$

with $V_\Sigma = \int_{\Sigma_2} \omega_2 \wedge \cdot_2 \omega_2$. The obstruction's unknown unit is thus expressed through three computable data: the internal cycle metric V_Σ , the gauge coupling g_6 , and the minimal charge e_6 . **A scope caveat (essential):** the “6” of F33.3A' is the *representation* dimension of the Y-sector field content — $\dim \Lambda^2(\mathbb{R}^{\wedge\{1,3\}}) = \dim \mathfrak{so}(1,3) = \dim(E \oplus B) = 6$ — and it does *not* by itself imply a *six-dimensional spacetime*. The reduction above is therefore stated in the safer *internal-fibre* form: $Y_6 = M_4 \times \Sigma_2$ is a total configuration bundle over the 4D base M_4 carrying a *rank-2 internal cycle* Σ_2 , not a Kaluza–Klein spacetime uplift. Whether the six field slots additionally *force* a genuine 6D metric is a separate question, deferred to ZS-F34. **Theorem F33.8D (conditional):** given the rank-2 internal-cycle ansatz, if (Σ_2, ω_2) are unique in the corpus geometry, g_6 is derived from the full Y spectrum, e_6 is fixed by large-gauge invariance, and the wrapped-brane sector is anomaly-free, then $\chi_- = e_-^2 / (4\pi^2 Z_-)$ is DERIVED. Establishing the ansatz and these data is the subject of ZS-F34, *The Six-Dimensional Charge Unit of the Z-Spin Three-Form*; in F33 the reduction is registered as DERIVED-CONDITIONAL under that ansatz, turning the open “absolute χ_- ” into a concrete, bounded computation. **F33-G8D:** if the field-representation “6” is used to assert a 6D spacetime without an independent metric, or the reduction yields a V_Σ , g_6 , or e_6 inconsistent with the corpus geometry, the route is RETRACTED.

§10. Conclusion

ZS-F33 v1.8 is a conditional UV reduction, not a completion, and says so. Its load-bearing advances are now each tied to an explicit computation, with statuses set at what is actually proven. The odd carrier is localized by an *imported* even-dimensional torsion no-go with a conditional Z-application (F33.1A/B); odd-dimensional uniqueness is *CLOSED-NEGATIVE*, refuted by the 2D seam-parity-restricted determinant torsion $\ln T_{2-} = \ln 4 \neq 0$ (F33.1C). The transgression holonomy $\theta_Z = \omega$ is DERIVED-under-WPL through a flat Wilson polar-line $L_- W$ ($c_1 = 0$, holonomy $e^{\wedge(i\omega)} = \lambda^* / |\lambda^*|$), kept strictly separate from the degree-1 theta line $\Theta_- Z$ ($c_1 = 1$, fibre flux) — repairing the v1.6 conflation of a flat bundle with a degree-1 one. The physical fibre itself is closed: the Wilson–Koenigs orbit-space theorem makes $\mathbb{C}^* / \langle \lambda^* \rangle = E_{\lambda^*}$ the orbit space of the F0 Wilson dynamics (DERIVED-under-Orbit-Equivalence), with the seam eigenvector $(1, -i)/\sqrt{2}$ selecting the λ^* branch.

Gauge invariance and seam parity are shown *not* to isolate the odd sector — a BRST-Only Isolation No-Go (F33.4A). The v1.6 claim that $\int F_4$ is a nontrivial local $H^{04}(s|d)$ class is *retracted* (it is d-exact); the corrected mechanism is a two-four-form sequestering action with the corpus $F_\pm$, whose abelian-ghost-tower BV master equation reduces to $d^2 = 0$ (DERIVED-CONDITIONAL) and whose gluing functions are pinned by the exponential-uniqueness theorem $\sigma = e^{\wedge x}$ (F33.4B). The centerpiece, F33.5, is now a *complete cellular-seam classification*: the truncated-octahedron BCC T^3 quotient has 48 cellular automorphisms and 20 seam involutions, and the seam-odd torsion image is the finite signed

half-lattice $\{-3, 0, 1, 3/2, 2, 5/2\} \cdot \ln 2$ — disjoint from the back-solved target, which lies in neither the ordinary non-negative subdeterminant lattice (nearest 8.148) nor the torsion image. The register dimensions are independently re-derived: the bivector–complexification consistency $2(d-1) = d(d-1)/2$ forces $d = 4$, $X = 3$, $Y = 6$, recovering $Q = 11$ without assuming it (F33.3A').

On the absolute scale we are deliberately conservative. The three registered routes are CLOSED-NEGATIVE-under-R1–R3, and the Charge-Unit Obstruction explains the structural reason: flux integrality fixes the flux number k but not the dimensionful unit $\chi_- = e^{-2}/(4\pi^2 Z_-) \propto e^{-2}/Z_-$, so (A, Q) and topology alone cannot determine χ_- (F33.8). This is *not* an impossibility theorem: the obstruction names exactly the missing datum, and the internal charge-unit reduction (F33.8D, developed in the companion ZS-F34) shows how a rank-2 internal-cycle compactification would supply it, giving $\chi_- = e^{-2}/(4\pi^2 Z_-)$ as DERIVED-CONDITIONAL on that ansatz. The honest content of conditional reduction is intact: the *structure* — the Wilson phase, the elliptic carrier and its orbit space, the BV sequestering, the complete spectral classification, the dimensional re-derivation — is derived, while the *absolute scale* awaits the one UV datum the obstruction isolates. ω stays distinct from the gauge A (28×). The cosmology is the frozen $w = -1$ branch, consistent with ZS-A28/A29; the dynamical- w branch is CLOSED-NEGATIVE under the minimal action and a NON-CLAIM otherwise. $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

Acknowledgements & Code Availability

This revision integrates an internal peer review that corrected three numerical errors (the half-weight $11 \ln 2 + \frac{1}{2} \ln 3 = 8.174 \neq 8.190$; the geometric mean $8.218 \neq 8.233$; the coexact spectrum $\{8^3, 12^1\}$), reclassified over-assigned statuses, weakened the title from “completion” to “conditional reduction,” flagged the A28/A29 cosmology conflict, and prescribed the BRST no-go pivot and the cellular-torsion construction. All numerical checks — ω and $\omega^2/2$, the Factorized-Determinant vev, the eighth-power blind benchmark, the octahedral cellular involution and its seam-odd split, and the 276.6 collapse — were performed with standard open-source libraries (Python/NumPy/mpmath, seed 437) and are reproduced in Appendix A and in `zs_f33_verify_v1_8.py`, which is structured so that no observed or back-solved value enters the computation of $C_{\text{odd}}^{\text{sp}}$. $C_{\text{odd}}^{\text{sp}}$ is not derived here; $\rho_\Lambda = \Delta E(\omega)$ is a falsification target. PROVEN / IMPORTED-PROVEN results rest on standard theorems (Ray–Singer; Cheeger–Müller; Cheeger–Simons / Bär–Becker; Barnich–Brandt–Henneaux; Henneaux–Knaepen–Schomblond; Witten; Bonati et al.; Kaloper–Padilla; Klinkhamer–Volovik; Chandrasekaran–Longo–Penington–Witten).

Appendix A. Verification Ledger

Verification reclassification (v1.8). The script is now *standalone* — it builds the BCC T^3 quotient and the full 48/20 classification inline, with *no sidecar files*. The suite is split into four kinds, and only ASSERT-COMPUTED counts toward a “pass”. **ASSERT-COMPUTED** (19): assertions that can fail — incidence matrices, $\partial_1 \partial_2 = 0$, Betti numbers, the full Δ_1 spectrum, the 48-automorphism / 20-involution classification and its torsion image, the full 11×11 Wilson map with $P_Z W P_Z = W_Z$, the Wilson polar phase, the Z -block eigenvalues and the J_Z branch selection, the 2D parity torsion computed from the Fourier spectrum ($\ln T_- = \ln 4$), the bivector identity $d = 4$, and the $\chi_- = e^{-2}/Z_-$ sign.

IDENTITY-REPORTED (4): definitional identities — $q = e^{2\pi i \tau} = \lambda^*$, $\omega = \arg \lambda^*$, the dimensional identities. **IMPORTED-PROVEN** (1): the multiplicative Cauchy uniqueness $\sigma = e^x$ [18e], not a finite-grid check. **STRUCTURAL-ASSUMPTION** (3): declared interpretations excluded from the pass

count — the orbit-equivalence postulate $O(w)=O(\lambda^*w)$, the corpus-F- sequestering correspondence, and the canonical-kinetic selection. The retracted $\int F_4$ booleans and the four declarative $c_1/b_2 \square/w$ booleans are removed entirely.

Table A1. Numerical regressions (the v1.8 standalone verification suite; reclassified into ASSERT-COMPUTED / IDENTITY-REPORTED / IMPORTED-PROVEN / STRUCTURAL-ASSUMPTION), from locked inputs and standard constants only. (TO = TO quotient; BQ = Berry–Quillen / Koenigs; Θ/Θ' = theta line; KP = Kaloper–Padilla; KT = Koenigs torus; BV = Batalin–Vilkovisky; NOME = nome $q = \lambda^*$; AN = anti-numerology; A26 = present epoch; ET = equivariant torsion 2D/3D; AUD = integrity audit; CLS = complete cellular-seam classification; WPL = Wilson polar-line; FG = flux-gluings uniqueness; F1C = F33.1C Fourier; CUO = charge-unit obstruction; WK = Wilson–Koenigs orbit space; Y6 = bivector consistency.)

#	Check	Result
1	$\omega = \arg \lambda^* (0 < \omega < \pi)$	2.2592495539
2	$\omega^2/2$	2.5521042734
3	$\kappa = -\ln \lambda^* $	0.1148346250
4	$C_M^{\wedge sp} = 11 \ln 2 + \ln 3 = \ln 6144$ (coexact $\{8^3, 12^1\}$)	8.7232312748
5	exact-block $\ln \det \{4^3, 6^2\} = 8 \ln 2 + 2 \ln 3$	7.7424020218
6	nearest half-weight $11 \ln 2 + \frac{1}{2} \ln 3$ (NOT 8.190)	8.1739251306
7	arithmetic mean of (5),(4)	8.2328166483
8	geometric mean of (5),(4) (NOT 8.233)	8.2181966839
9	Reidemeister $S_cl = 35\pi/3$	36.6519142919
10	Ray–Singer $\gamma_CW \cdot C_M^{\wedge sp} = (38/9)(11\ln 2 + \ln 3)$	36.8314209382
11	$v = \bar{M}_P \cdot \exp(-36.831)$ (Factorized Determinant)	245.93 GeV
12	blind $\rho_A^{\wedge \{1/4\}} = v^2/\bar{M}_P$	2.484×10^{-5} eV
13	$C_odd^{\wedge sp}$ target = $34.58/(38/9)$ (back-solved)	8.1901
14	$b_2\omega^2$ (SU(3) $b_2 = -0.0216$); $1+b_2\omega^2$	-0.110; 0.890
15	276.6 collapse $32\pi^2/(b_Z A)$, $b_Z = 14.25$	276.7 (= epoch)
TO	TO/BCC T^3 quotient (V,E,F,C); χ ; Betti	(6,12,7,1); 0; (1,3,3,1)
TO	$\partial_1\partial_2 = 0$; rank ∂_1, ∂_2	True; 5, 4
TO	spec $\Delta_1 = \partial_1^T\partial_1 + \partial_2\partial_2^T$ (corpus ZS-Q3)	$\{0^3, 4^3, 6^2, 8^3, 12^1\}$
TO	$J_0^2=J_1^2=J_2^2=I$; $J_0\partial_1=\partial_1J_1$; $J_1\partial_2=\partial_2J_2$; $[J_p, \Delta_p]=0$	all True
TO	seam-odd blocks $\Delta_0 -, \Delta_1 -, \Delta_2 -$	$\{4^3\}, \{4^3\}, \emptyset$
TO	equivariant torsion $\ln T_- = \frac{1}{2}\Sigma(-1)^p \ln \det'(\Delta_p _-)$	$-3 \ln 2 = -2.0794$
TO	$8.190 \in \mathcal{L} = \mathbb{Z}_{\geq 0} \ln 2 \oplus \mathbb{Z}_{\geq 0} \ln 3$?	NO (nearest 8.1479)
BQ	i-tetration orbit rotation number per step	2.25925 (= ω , 5e-6)
BQ	Koenigs monodromy det phase = $\arg \lambda^*$	2.25925 = ω (exact)
BQ	engineered determinant-line Berry phase $\oint A_B$	2.25925 (= ω , 4e-10)
BQ	ω vs gauge A (distinct bundles)	$\omega/A = 28.21$
Θ	c_1 of det-line on $\text{Jac}(Z=T^2)$: winding of $\arg \theta_1 / 2\pi$	1 (theta divisor)
Θ	c_1 matches a_Z normalization $\int_Z \text{curv} = 1$	True
KP	corpus 4-form $F=dA_3^-$, $\rho_Z=0 \cong$ Kaloper–Sorbo 4-form	structural match
KP	IR shift $\delta\chi_-/\chi_- \sim \langle R \rangle/\bar{M}P^2$ (dim 6), RG-stable	3.5×10^{-121}
KP	$\Omega_A/\Omega_m = 2e^A$ vs KP residual $O(1)$	2.167

#	Check	Result
KT	Koenigs $\tau = \log(\lambda^*)/2\pi i = (\omega + i\kappa)/2\pi$	$0.3596 + 0.0183 i$
KT	$\text{Im } \tau = \kappa/2\pi > 0$ (valid elliptic curve)	$0.01828 > 0$
Θ'	c_1 of theta line for the derived τ	1
BV	$F_4 = dA_3^-$ gauge variation sF_4	0 (gauge-invariant)
BV	$\int F_4 = dA_3$ locally $\Rightarrow$ d-exact $\Rightarrow$ trivial in $H^{0,4}(S d)$	RETRACTED (body §5.2)
FORK	$E^*_{\text{need}}/v = \sqrt{(\rho_{\text{obs}}/\rho_{\text{blind}})} (= \sqrt{90})$	9.49
NOM E	$q = e^{\wedge\{2\pi i \tau\}}$ vs $\lambda^* : q - \lambda^* $	1.4×10^{-31} ($q = \lambda^*$)
NOM E	$j(\tau) = E_4^3/\Delta$ with $q = \lambda^*$	$1.06 \times 10^5 + 2.41 \times 10^5 i$
AN	e^ω near-miss for E^*/v (rejected, no mechanism)	9.58 vs 9.49 ($\approx 1\%$)
A26	$v_{\text{now}} = \ln(\tilde{M}P^4/\rho_{\text{obs}}) = \text{present epoch}$	276.64 (tautological)
ET	2D T^2 seam-odd torsion (even): $\ln T^-$ (all chain-map signs)	$\ln 4 = 1.386 (\neq 0)$
ET	2D T^2 full Poincaré duality $\det' \Delta_0 = \det' \Delta_2$	$20.337 = 20.337$
ET	2D T^2 τ -odd duality $\det' \Delta_0 _-$ vs $\det' \Delta_2 _-$ (fails)	$7.742 \neq 10.515$
ET	3D T^3 seam-odd torsion (odd): $\ln T^-$	$6 \ln 2 = 4.159$
AUD	ZS-M1 $z^* = i^\wedge(z^*)$ fixed-point residual	1.1×10^{-31}
AUD	register $Q = 11 = 3(X) + 6(Y) + 2(Z)$	$11 = 11$
AUD	nome $q = e^{\wedge\{2\pi i \tau\}} = \lambda^*$ (chain root $\rightarrow$ carrier)	$ q - \lambda^* < 10^{-30}$
CLS	$ \text{Aut}_{\text{cell}}(\text{BCC } T^3) $ (signed cellular automorphisms)	$48 = \text{O}_h $
CLS	order-2 cellular seam involutions (incl. identity)	20
CLS	seam-odd torsion image $\mathcal{T}_-$ over all involutions	$\{-3, 0, 1, 3, 2, 2, 5, 2\} \cdot \ln 2$
CLS	target $8.190 \in \mathcal{T}_- ?$ (max $ \mathcal{T}_- = 2.079$)	False
WPL	Wilson polar phase $\arg(\lambda^*/ \lambda^*) = \omega$	2.25925 (resid 10^{-12})
WPL	flat Wilson line $c_1(L_-W) = 0$; theta line $c_1(\Theta_-Z) = 1$	0 / 1 (distinct)
FG	flux-gluing $\sigma(x+y) = \sigma(x)\sigma(y)$, $\sigma(0)=1$, $\sigma'(0)=1 \Rightarrow \sigma = e^\wedge x$	exp (Cauchy unique)
FIC	2D Fourier $\ln \det'(\Delta_0 _-)$: 6 odd-mode pairs	7.742
CUO	Charge-Unit Obstruction $E_-k(\theta) = (e^{-2/2Z_-})(k + \theta/2\pi)^2$	$\chi_- = e^{-2/(4\pi^2 Z_-)} \propto e^{-2/Z_-}$
WK	Wilson Z-block W_-Z eigenvalues; $(1, -i)/\sqrt{2}$ is the λ^* eigenvector	λ^*, λ^* ; branch selected
WK	$\mathbb{C}^*/\langle \lambda^* \rangle$ free & properly discontinuous ($0 < \lambda^* < 1$); $\text{Im } \tau > 0$	E_{λ^*} torus, $\text{Im } \tau = 0.0183$
Y6	$2(d-1) = d(d-1)/2$ unique solution $d > 1$	$d=4 \Rightarrow X=3, Y=6$
Y6	four routes to 6: 3×2 , $C(4,2)$, 3_-J+3_-K , $(1,0) \oplus (0,1)$	all = 6

Computation TO (full quotient): the truncated octahedron (24 vertices, 36 edges, 14 faces) modulo $\Lambda = \langle (2,2,2), (4,0,0), (0,4,0) \rangle$ gives $(6,12,7,1)$ with the ZS-Q3 edge spectrum reproduced exactly. The central inversion descends to J_0, J_1, J_2 satisfying $J^2=I$, both chain maps, and $[J_p, \Delta_p]=0$ (errors $< 10^{-12}$). The seam-odd determinants are $\{4^3\}$ at $p=0,1$ and empty at $p=2$, giving $\ln T^- = -3 \ln 2$. Since $\{4,6,8,12\}$ factorize over $\{2,3\}$, all achievable log-determinants lie in $\mathcal{L} = \mathbb{Z}_{\geq 0} \ln 2 \oplus \mathbb{Z}_{\geq 0} \ln 3$ and $8.190 \notin \mathcal{L}$ — the involution-independent no-go. Computation BQ (no observed input): the nonlinear i-tetration orbit rotation number, the Koenigs-linearized monodromy determinant phase, and an engineered determinant-line Berry holonomy all equal ω , distinct from the gauge A . No back-solved constant enters either computation; the comparison to 8.190 is a separate post-run block.

Computation Θ (theta line): for $Z = T^2$ with modular parameter $\tau = i$, the winding of $\arg \theta_1(u|\tau)$ around the fundamental cell of $\text{Jac}(Z)$ is exactly 2π , so c_1 of the determinant line is 1 — the degree of the theta divisor — equal to the F33.2A fiber-character normalization $\int_Z \text{curv} = 1$. The seam-odd characteristic is the odd theta characteristic θ_1 (single zero at the origin). Computation KP (structural): the corpus four-form with $\rho_Z = 0$ matches the Kaloper–Sorbo 4-form term; the admissible mixed-cocycle shifts are $\langle R \rangle / \bar{M}P^2 \sim 10^{-121}$ and $(\langle R \rangle / \bar{M}P^2)^2 \sim 10^{-241}$, radiatively stable under the KP sequestering constraint.

Computation KT (Koenigs torus): the multiplier $\log \lambda^* = -\kappa + i\omega$ gives $\tau = \log \lambda^* / (2\pi i) = (\omega + i\kappa) / 2\pi = 0.3596 + 0.0183 i$ with $\text{Im } \tau > 0$, so the Koenigs quotient $\mathbb{C}^* / \langle \lambda^* \rangle$ is a genuine elliptic curve and $Z = T^2$ is derived. Computation BV: $F_4 = dA_3^-$ is gauge-invariant under the reducible symmetry, and $\int F_4$ is a closed non-exact top-form — an admissible $H^{0|4}(s|d)$ cocycle. Computation FORK: the enhancement needed to lift the blind 2.48×10^{-5} eV to the observed 2.24×10^{-3} eV is $E^*_{\text{need}}/v = \sqrt{90} = 9.49$, which is not derived from (A, Q) and is declined as numerology (F33-G11).

Computation NOME: the nome $q = e^{2\pi i \tau} = \lambda^*$ to 10^{-31} — the elliptic-curve complex structure is the i -tetration multiplier itself; $j(\tau)$ is then a determined function of λ^* with no forced (A, Q) relation.

Computation AN (anti-numerology): the $\sqrt{90}$ enhancement is numerically close to $e^\omega = 9.58$ ($\approx 1\%$) but e^ω of a rotation angle has no mechanism and is rejected; all other matches need fitted exponents.

Computation A26: $v_{\text{now}} = \ln(\bar{M}P^4 / \rho_{\text{obs}}) = 276.64$ reproduces the present-epoch constant, confirming Branch B is tautological.

Appendix B. Falsification Gates

Gate	Layer	Condition
F33-G1	Mathematical	A standard closed even-dim or gauge-mode carrier nonetheless yields $\rho_{\Lambda, Z} \neq 0$.
F33-G5	Mathematical	The complete cellular-seam classification (48 automorphisms / 20 involutions) is shown incomplete, or a 21st order-2 cellular action is exhibited.
F33-G2'	Mathematical	The Z -fibration degenerates ($\gamma_Z \subset Z$), so $C_3^- = a_Z \wedge \Omega_Z$ vanishes.
F33-G2''	Theoretical	Berry–Quillen holonomy $\neq e^{\{i\omega\}}$, or a_Z carries only the gauge A; $\theta_Z = \omega$ RETRACTED.
F33-G3 / G5	Meta / zero-parameter	$g^2(\bar{M}_P)$, b_0 , the portal coefficient, or the odd scale is an external or inverted-observed input.
F33-G4	Non-perturbative	A sequestering failure lifting $ \delta \rho_{\Lambda} / \rho_{\Lambda} $ above ε_{rad} (mixed cocycles survive in the IR).
F33-G6	Meta / anti-numerology	$C_{\text{odd}}^{\text{sp}}$ obtained by fitting a non-integer weight to 8.190 rather than from the seam-odd spectrum.
F33-G7	Physical	$\varepsilon_N(\omega) > 1\%$ but the cumulant series is truncated at b_2 for cosmology.
F33-G8	Observational	Computed ρ_{Λ} inconsistent with data (no retuning); Branch B claimed with $K(q)$ free yet zero-parameter asserted.
F33-G9	Meta / anti-numerology	χ_- or $C_{\text{odd}}^{\text{sp}}$ claimed parameter-free using observed ρ_{Λ} , a chosen scale, base-10, a half-weight, or a back-solved exponent.
F33-G10	Meta / zero-parameter	The KP non-spectral route claimed to fix ρ_{Λ} parameter-free while the 4-form flux is left free or not matched to $\Delta E_{\text{--}}(\omega)$.
F33-G2theta	Mathematical	The corpus forces a carrier whose Koenigs quotient is not genus-1, or fixes $\tau \neq (\omega + i\kappa) / 2\pi$ — voids the $c_1=1$ identification and $\theta_Z = \omega$.

Gate	Layer	Condition
F33-G11	Meta / anti-numerology	The value-closure fork resolved by fitting $\sqrt{90}$ or 90 to A, Q rather than deriving the enhancement.
F33-G1C	Mathematical	The 2D counterexample $\ln T_- = \ln 4 \neq 0$ is shown to be in error (restoring odd-dim uniqueness), or ZS-Q3 is shown not to live on the 3D X-ambient complex (breaking the assignment).
F33-G2orb	Mathematical	The F0 Wilson Z-block does not have λ^* as a simple eigenvalue, or the seam chirality $(1,-i)/\sqrt{2}$ does not select the λ^* branch; the orbit-space identification $\mathbb{C}^*/(\lambda^*) = E_{\lambda^*}$ is RETRACTED.
F33-G8D	Theoretical	The field-representation 6 is used to assert a 6D spacetime without an independent metric, or the rank-2 internal-cycle reduction yields a V_{Σ} , g_6 , or e_6 inconsistent with the corpus geometry.

Appendix C. Anti-Numerology Audit of $C_{\text{odd}}^{\text{sp}}$ (corrected)

Candidate	Finding	Verdict
$b_0 = 14.25$ (fourth power)	$32\pi^2/(b_{\text{Z}} A) = 276.6 = \text{present epoch (A26)}$	CLOSED-NEGATIVE (tautology)
exact block $\{4^3, 6^2\} = 7.742$	corpus-native; misses target	$\neq 8.190$
coexact $\{8^3, 12^1\} = 8.723$	corpus-native (= C_{M}^{sp}); misses	$\neq 8.190$
seam-odd equivariant torsion = -2.079	canonical central inversion (full complex)	$\neq 8.190$
$8.190 \in \mathcal{L} = \mathbb{Z}_{\geq 0} \ln 2 \oplus \mathbb{Z}_{\geq 0} \ln 3$?	Q3 eigenvalues factor over $\{2,3\}$; $8.190 \notin \mathcal{L}$	CLOSED-NEGATIVE (decisive)
nearest half-weight $11 \ln 2 + \frac{1}{2} \ln 3 = 8.174$	$\frac{1}{2}$ not spectrum-derived	numerology if fitted
arith. mean 8.233 / geom. mean 8.218	not derived combinations	no mechanism

Two lattices, kept distinct (essential). The “ $\in \mathcal{L} = \mathbb{Z}_{\geq 0} \ln 2 \oplus \mathbb{Z}_{\geq 0} \ln 3$ ” row refers only to the *ordinary subdeterminants* of ZS-Q3 — the non-negative integer lattice in which the spectral $C_{\text{odd}}^{\text{sp}}$ route would live (nearest point $7 \ln 2 + 3 \ln 3 = 8.148$). It is *not* the same set as the seam-odd determinant-torsion image $\mathcal{T} = \{-3, 0, 1, 3/2, 2, 5/2\} \cdot \ln 2$, a *signed half-lattice* containing negative and half-integer multiples (e.g. the canonical $-3 \ln 2$). The target 8.190 misses *both* — $\mathcal{L}$ (ordinary, non-negative) and $\mathcal{T}$ (signed torsion) — by independent margins. Conflating the two was the v1.6 error this version corrects; they are reported as separate scopes throughout (§6.4).

The decisive entry is the last: because every Q3 eigenvalue factorizes over $\{2,3\}$, every achievable spectral / equivariant log-determinant lies in the integer lattice $\mathcal{L}$, and 8.190 is provably outside $\mathcal{L}$. No seam involution can produce it; the spectral route is CLOSED-NEGATIVE. The observed value is never an input to the computation.

Appendix E. Issue-Tree Exploration Record (v1.8)

v1.8 integrates two further peer reviews whose verdict was “direction and central computation succeeded, but document/code integration unfinished.” The issue-tree separates the closable items from the deferred ones.

E.1 Long list → issue list

Step 0 (long list, 7): **A** the inverted χ_- formula (central theorem); **B** the non-standalone verifier (sidocar dependence); **C** the v1.6/v1.7 remnants in §10/§2.3/counts/filenames; **D** closing physical Z via Wilson

orbit space; **E** re-deriving (3,6,2) independently of Q; **F** the F33.3B KV over-flattening and the BV over-claim; **G** the absolute- χ - 6D route. Step 1 (issue list, 4): **1**=[A+B+C] the must-fix integration, **2**=[D] Wilson–Koenigs orbit space, **3**=[E+F] dimensional re-derivation, KV split, BV demotion, **4**=[G] the internal-cycle reduction (deferred to F34).

E.2 Traversal and resolutions

Issue	Resolution (computation)	Outcome
1A	$E_k = (e^{-2/2Z}) (k + \theta/2\pi)^2 \Rightarrow \partial^2 E / \partial \theta^2 = e^{-2/(4\pi^2 Z)}$	$\chi \propto e^{-2/Z}$ - (corrected, all sites)
1B	standalone builder reproduces (6,12,7), Δ_1 , 48/20 with no sidecar	verifier self-contained, 17/17
1C	rewrote §10; removed §2.3 remnants; fixed v1.7 filenames; 31-result counts	remnants cleared
2	$W_Z w = \lambda^* w$ on $\mathbb{C}^*$; free+properly disc.; $(1, -i)/\sqrt{2} \rightarrow \lambda^*$	$Z = E_{\lambda^*}$ DERIVED-under-Orbit-Equiv.
3a	$2(d-1) = d(d-1)/2 \Rightarrow d=4$; four routes to 6; split math/phys	$X=3, Y=6$ math PROVEN; phys COND
3b	KV split; BV demoted to DERIVED-CONDITIONAL ($S_{BV} \rightarrow d^2=0$)	statuses honestly lowered
4	$Y_6 = M_4 \times \Sigma_2$ internal-cycle reduction gives $\chi = e^{-2/(4\pi^2 Z)}$	F33.8D DERIVED-CONDITIONAL $\rightarrow$ F34

Step 4 (convergence): two passes; the second confirmed the orbit-space and bivector theorems are genuinely independent (the bivector route uses only $Z = 2$ and Lorentz Λ^2 , never $Q = 11$), that the χ -correction propagated to all sites (§9.4, §9.5, ledger, verifier), and that the representation “6” and a 6D spacetime are now explicitly separated. Converged. **Step 5 (score)**: two OPEN items are closed — physical $Z = E_{\lambda^*}$ (DERIVED-under-Orbit-Equivalence) and the register dimensions (math PROVEN) — one central sign error is fixed, two over-claims (BV master, Y-6D uplift) are lowered to their honest status, and the verifier is reproducible. The remaining absolute- χ - gap is the bounded F33.8D reduction handed to ZS-F34. The firewall is unchanged: ρ_{obs} enters only the post-run zone.

Appendix F. Integrity Audit (v1.0–v1.8)

v1.8 applies the nine-step verification protocol (§3.1–3.9) to the cumulative 31-result ledger and to the cross-version dependency chain. All checks pass; the table summarizes them.

Step	Check	Result
§3.1 parameters	Only (A, Q) = (35/437, 11) enter; no fitted constant; ρ_{obs} never an input	PASS
§3.2 chain	ZS-M1 $z^* = i^{\wedge}(z^*) \rightarrow \lambda^*$, κ , ω , $\omega^2/2$, C_M^{sp} , v , τ , $q = \lambda^*$ all flow from one root	PASS (resid 10^{-31})
§3.2 dimensions	Z 2D (Koenigs torus, θ_Z) vs X 3D (BCC T^3 , ZS-Q3) not conflated; $Q = 11 = 3+6+2$	PASS
§3.3 data	Frozen $w = -1$ branch consistent with ZS-A28/A29 and DESI DR2 / Planck	PASS
§3.4 legend	Every ledger tag is a legend value (legend expanded in v1.7); statuses consistent across v1.0–v1.8	PASS
§3.5 gates	Falsification gates F33-G1...G12, G1C, G2WPL, G2orb, G8D present and current	PASS
§3.6 references	PROVEN imports (Ray–Singer, Cheeger–Müller, Koenigs/Milnor, Quillen, BBH, Kaloper–Padilla, Aczél) cited	PASS

Step	Check	Result
§3.7 structure	Epistemic legend, ledger, cross-version notes, per-cluster sections present	PASS
§3.9 firewall	Anti-numerology: e^ω near-miss rejected; B3 value CLOSED-NEGATIVE-under-R1–R3; ρ_{obs} only in fenced post-run zone	PASS

Cross-version status reconciliation. Statuses move only with a recorded driver: $\theta_Z = \omega$ rose OPEN $\rightarrow$ DERIVED-CONDITIONAL (v1.2) $\rightarrow$ DERIVED (v1.4) and was then *corrected* to DERIVED-under-WPL (v1.7) after the c_1 error was found; F33.4B rose to DERIVED (v1.4) and was *lowered* to DERIVED-CONDITIONAL with the $\mathbb{F}_4$ class retracted (v1.7); the vacuum *value* fell to CLOSED-NEGATIVE-under-R1–R3 (v1.5/v1.7); F33.1C uniqueness is CLOSED-NEGATIVE (v1.7, counterexample); and the spectral no-go was rescoped to the complete cellular-seam classification (v1.7). In v1.8: physical $Z = E_{\lambda^*}$ was closed to DERIVED-under-Orbit-Equivalence via the Wilson–Koenigs orbit space; the bivector consistency was split into F33.3A'-math (PROVEN) and F33.3A'-phys (DERIVED-CONDITIONAL); the BV master action was lowered from PROVEN to DERIVED-CONDITIONAL once the explicit abelian ghost-tower antibracket was found to be exhibited only as a reduction to $d^2 = 0$; and the inverted χ_- was corrected to $e^{-2/(4\pi^2 Z_-)}$ consistently through §9.4, §9.5, the ledger, and the verifier. No earlier result was silently reversed: each transition is recorded in the Version History with its driver. The single dimensional subtlety — that the seam-odd determinant of §6 is computed on a 3D complex while the mediation angle lives on a 2D one — is resolved by the sector split (Appendix audit row §3.2): they are different invariants on different register sectors, X (3D) and Z (2D), and are never interchanged.

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Version History

v1.8 (May 2026): Closure and integration update from two further peer reviews. Closed two open items, corrected a sign error in the central theorem, and synchronized statuses. (1) Physical fibre: added the Wilson–Koenigs orbit-space theorem — the F0 Wilson Z-block acts as $w \mapsto \lambda^* w$ on $\mathbb{C}^*$, a free properly-discontinuous action whose quotient $\mathbb{C}^*/\langle \lambda^* \rangle = E_{\lambda^*}$ is the orbit space; “ $Z = E_{\lambda^*}$ ” is raised to DERIVED-under-Orbit-Equivalence (PROVEN as the orbit space; the observable orbit-invariance is the one remaining postulate), with the seam eigenvector $(1, -i)/\sqrt{2}$ selecting the λ^* branch (§3.2.4). (2) Register dimensions: added the bivector–complexification consistency $2(d-1) = d(d-1)/2 \Rightarrow d = 4, X = 3, Y = 6$, re-deriving $Q = 11$ without assuming it via four routes to 6, split into F33.3A'-math (PROVEN) and F33.3A'-phys (DERIVED-CONDITIONAL) (§4.1). (3) F33.3B split into IMPORTED-PROVEN-within-model / NON-CLAIM / DERIVED-CONDITIONAL / OPEN-executable (§4.3). (4) Charge-Unit Obstruction: corrected the inverted formula to $\chi_- = e^{-2/(4\pi^2 Z_-)} \propto e^{-2/Z_-}$ consistently through §9.4, §9.5, the ledger, and the verifier; added the internal charge-unit reduction F33.8D under an explicit rank-2 internal-cycle ansatz, separating the representation-dimension “6” from a 6D spacetime, pointing to ZS-F34 (§9.5). (5) Lowered the BV master action from PROVEN to DERIVED-CONDITIONAL, giving the explicit abelian ghost-tower S_{BV} whose master equation reduces to $d^2 = 0$ (§5.2). Editorial closure: rewrote the §10 conclusion (which carried v1.3/v1.6 text), removed the “Open Gate / what remains OPEN” remnants in §2.3, separated the ordinary subdeterminant lattice from the signed torsion image in Appendix C, updated the gates (G2orb, G8D) and Appendix F to v1.8, fixed the v1.7 verifier filenames, unified all counts (31-result ledger), and made the verification script standalone — it builds the BCC T^3 quotient and the 48/20 classification inline with no sidecar files (17/17 ASSERT PASS), computes the 2D parity torsion from the Fourier spectrum, and reclassifies Cauchy uniqueness as IMPORTED-PROVEN. **v1.7 (April 2026):** Status-correction and classification update integrating a detailed peer review. Lowered three over-claimed load-bearing statuses and retracted one incorrect cohomology claim, while adding four computation-backed results. (1) $\theta_Z = \omega$: the v1.3–v1.6 identification was flawed (a flat connection has $c_1 = 0$ and cannot be the degree-1 theta line); v1.7 separates a flat Wilson polar-line L_W ($c_1 = 0$, holonomy $e^{\{i\omega\}} = \lambda^*/|\lambda^*|$ from the polar unitary part of the linearized mediation operator) from the theta line Θ_Z ($c_1 = 1$, fibre flux), giving $\theta_Z = \omega$ as DERIVED-under-WPL and demoting the engineered Berry loop to a realizability check (§3.2). (2) F33.4B: retracted the claim that $[F_4]$ is a nontrivial local $H^{04}(s|d)$ class (it is d-exact) $\rightarrow$ OPEN-GLOBAL;

wrote the two-four-form action $S_{\text{ZS-seq}}$ with the corpus $F_{\pm}$, proved the BV master equation, and closed the gluing free-function by the exponential-uniqueness theorem $\sigma = e^x$ (Cauchy) (§5.2). (3) F33.5: replaced the over-scoped non-negative-lattice no-go (which contradicted the canonical $-3 \ln 2$) with the complete cellular-seam classification — 48 automorphisms, 20 involutions, torsion image $\{-3, 0, 1, 3/2, 2, 5/2\} \cdot \ln 2$, target disjoint — exact for all cellular involutions on this complex (§6.4). (4) F33.1C: the 2D parity-restricted torsion $\ln T = \ln 4 \neq 0$ is a counterexample, so odd-dimensional uniqueness is CLOSED-NEGATIVE, with terminology corrected to “seam-parity-restricted determinant torsion” (§2.3). Added the canonical compact 3-form branch ($b_2 \square = 0$, PROVEN-under-canonical-action, §7), closed the dynamical-w Outlook as CLOSED-NEGATIVE under the minimal action (§8.2), and proved the Charge-Unit Obstruction (topology fixes the flux number, not the dimensionful unit; the strongest value terminus, §9.4). Reclassified the verification suite into ASSERT-COMPUTED / IDENTITY-REPORTED / STRUCTURAL-ASSUMPTION, removing the declarative passes. Fixed editorial errors (ledger count, §10 and code-availability version strings, the F33-G1C duplicate, the $\log \det'$ notation, the $\Omega_{\Lambda}/\Omega_m$ “same order” label, and the legend coverage). New title; references [18e] (Aczél) added. **v1.6 (April 2026):** Integrity-audit update. Tested the F33.1C closure (odd-dimensional carrier uniqueness): since the seam is the central inversion, orientation-preserving in even dimensions, one might hope the seam-odd equivariant torsion vanishes on the 2D even carrier. Explicit computation REFUTES this — on a 2D T^2 the τ -odd Poincaré self-duality fails ($|\det' \Delta_0| = 7.742 \neq |\det' \Delta_2| = 10.515$ under every chain-map sign), giving $\ln T = \ln 4 \neq 0$, while the 3D T^3 gives $6 \ln 2$ — so the even-dimensional equivariant torsion is genuinely nonzero (confirming v1.1's Problem B) and F33.1C is CONFIRMED-OPEN (§2.3.1). Recorded the sector assignment (odd vacuum on the 3D X-ambient via ZS-Q3, θ_Z on the 2D Z Koenigs torus) as DERIVED-from-structure (§2.3.2). Added a full integrity audit (Appendix F): the ZS-M1 z^* dependency chain (residual 10^{-31}), the Z-2D / X-3D sector separation ($Q = 11 = 3+6+2$), and the nine-step protocol §3.1–3.9, all passing. Added the equivariant-torsion and audit computations to Appendix A, gate F33-G1C, and one ledger row (25 total). **v1.5 (March 2026):** Deep-exploration (issue-tree) update executing the three v1.4 next-steps. F33.2B: proved the elliptic-curve nome $q = e^{\{2\pi i \tau\}} = \lambda^*$ (the carrier's complex structure is the i -tetration multiplier), confirming cross-version consistency with ZS-M and F27 (§3.2.3); reported $j(\tau)$ without forcing an (A, Q) relation. F33.8: completed the absolute-value question as CLOSED-NEGATIVE by exhausting three independent routes — the spectral lattice no-go (v1.2), Branch A whose $\sqrt[90]$ enhancement is not derivable from (A, Q) (the e^{ω} near-miss is rejected as the exponential of a rotation angle with no mechanism, F33-G11), and Branch B which gives $\rho_{\Lambda} = \bar{M}_P e^{\{-276.6\}} = \rho_{\text{obs}}$ at the right magnitude but tautologically since the present epoch is defined by ρ_{obs} (§9.3). Added the nome, $j(\tau)$, anti-numerology and present-epoch computations to Appendix A, two ledger rows (24 total). **v1.4 (March 2026):** Deep-exploration (issue-tree) update executing the three v1.3 next-steps. F33.2B: derived the genus-1 carrier itself — the Koenigs quotient $\mathbb{C}^*/\langle \lambda^* \rangle$ of the i -tetration multiplier is an elliptic curve with $\tau = (\omega + i\kappa)/2\pi$ fixed by (A, Q) — removing the v1.3 structural input, so $\theta_Z = \omega$ is now DERIVED outright (§3.2.2). F33.4B: established BV-admissibility of the Kaloper–Sorbo term ($F_4 = dA_3^-$ gauge-invariant, $\int F_4$ a nontrivial $H^{04}(s|d)$ cocycle), promoting the local IR sequestering to DERIVED (§5.2). F33.8: attempted the B3 value-closure and found the parameter-free-vs-magnitude fork (blind value $90\times$ short; reconciliation needs $\sqrt[90]$ in E^* , declined as numerology), so the value stays OPEN with the obstruction now precise (§9.3). Added the Koenigs-torus, BV, and fork computations to Appendix A, gates F33-G11 and the revised G20, the Milnor reference, and four ledger rows (22 total). **v1.3 (March 2026):** Deep-exploration (issue-tree) update executing the three v1.2 next-steps. F33.2B: on the genus-1 carrier Z

$= T^2$, constructed the determinant line over $\text{Jac}(Z)$ as the theta line bundle and verified $c_1 = 1$ (winding of $\arg \theta_1$) = the F33.2A fiber-character normalization $\int_Z \text{curv} = 1$, identifying a_Z with the Quillen/BGS connection and the seam-odd sector with the odd theta characteristic — promoting $\theta_Z = \omega$ from DERIVED-CONDITIONAL to DERIVED (structural input: $Z = T^2$, forced by $\omega \neq 0$). F33.4B: identified the corpus four-form ($\rho_Z = 0$) with the Kaloper–Sorbo / Kaloper–Padilla sequestering 4-form, making the IR radiative-stability bound RG-stable via a PROVEN external mechanism. F33.8 (new §9.2): identified the KP sequestering global constraint as the non-spectral central normalizer B3 requires (HYPOTHESIS-strong), with value-closure open since the 4-form flux is free. Added the theta-line and KP computations to Appendix A, gates F33-G10 and F33-G2theta, the Quillen/Mumford references, and two ledger rows (20 total). **v1.2 (March 2026)**: Deep-exploration (issue-tree) update. Executed the three v1.1 next-steps. F33.5: built the full truncated-octahedron BCC T^3 quotient explicitly (∂_1, ∂_2 reproducing $\Delta_1 = \{0^3, 4^3, 6^2, 8^3, 12^1\}$ and Betti $(1, 3, 3, 1)$), verified the canonical central-inversion seam involution on every degree ($J^2=I$, both chain maps, $[J_p, \Delta_p]=0$), computed the seam-odd equivariant torsion $\ln T = -3 \ln 2$, and proved the involution-independent lattice no-go ($8.190 \notin \mathbb{Z} \geq 0 \ln 2 \oplus \mathbb{Z} \geq 0 \ln 3$) — closing the spectral route to the vacuum scale as CLOSED-NEGATIVE and strengthening A28. F33.2B: computed the i-tetration orbit rotation number, the Koenigs monodromy determinant phase, and an engineered determinant-line Berry holonomy (all $= \omega$, distinct from A), promoting $\theta_Z = \omega$ from OPEN to DERIVED-CONDITIONAL. Branch B kept OPEN (Outlook). Added Appendix E (issue-tree record), the Koenigs and Bismut–Gillet–Soulé references, and three new ledger rows (18 total). **v1.1 (March 2026)**: Major revision after internal review. Title weakened from “UV Completion” to “Conditional UV Reduction.” Corrected three numerical errors ($8.174 \neq 8.190$; geometric mean $8.218 \neq 8.233$; coexact $\{8^3, 12^1\}$). Reclassified statuses into a 16-result Claim Ledger (no DERIVED/PROVEN-under-product over-assignment). Added a cross-version consistency table resolving the A28/A29 cosmology conflict and the A28 spectral-route retirement. Split F33.1 into 1A (IMPORTED-PROVEN even-dim no-go), 1B (DERIVED-CONDITIONAL Z-application; dropped the erroneous $\text{ind}^*(D_Z)=\beta_0=1$), 1C (OPEN equivariant uniqueness). Split F33.2 into 2A (PROVEN-under-product transgression; corrected the $1/2\pi$ normalization) and 2B (OPEN Berry–Quillen holonomy). Split F33.3 into 3A/3B/3C/3D with retitled “classification and benchmark.” Pivoted F33.4 to a BRST-Only Isolation No-Go plus an IR radiative-stability bound. Promoted F33.5 to the paper’s center with an explicit octahedral cellular seam involution (seam-odd exact block $= 4.159$) and an anti-fit firewall admitting three outcomes. Retitled F33.6 with a controlled-truncation gate. Split F33.7 into a frozen $w = -1$ branch (main) and a dynamical-q Outlook. **v1.0 (March 2026)**: Initial release; consolidated A31; superseded by v1.1. (A, Q, dim Z) = (35/437, 11, 2) LOCKED.